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"PolyGlotTalk: A Context-Aware Cascade Speech-to-Speech Translation Pipeline for Low-Resource Offline Deployment"

Submitted in partial fulfilment for the award of the degree of

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(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

Submitted by

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Signature of Student(s)

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It gives us great pleasure to present the preliminary project report on

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ABSTRACT

The "PolyglotTalk" project addresses the critical need for real-time, fully offline Speech-to-Speech Translation (S2ST) systems for linguistically diverse and resource-constrained environments in India. While cloud-based solutions such as Google Translate and DeepL offer high accuracy, their dependency on stable internet connectivity and high-end server infrastructure renders them impractical for rural deployment or secure, offline communication. This project proposes a four-thread cascade pipeline consisting of Audio Capture, Automatic Speech Recognition (ASR), Machine Translation (MT), and Text-to-Speech (TTS), designed for hardware-adaptive offline deployment.

The primary technical contribution of this work is the implementation of a **rolling context window mechanism**. This deque-based prefix injection strategy addresses a common failure mode in streaming translation: loss of grammatical consistency and repetitive phrasing across audio chunk boundaries. Benchmarking on the project testbed (AMD Ryzen 7 7700X, and NVIDIA RTX 4060 Desktop GPU, WSL2 Ubuntu 22.04) demonstrates that the selected Argos Translate model achieves a corpus BLEU score of **0.5328** on 25 IIT Bombay English-Hindi sentence pairs, outperforming MarianMT (opus-mt-en-hi), which scored **0.2706** under identical conditions.

On the ASR stage, the base.en faster-whisper model achieved a mean WER of **0.0682** across 500 LibriSpeech clips, with a mean per-clip latency of **0.124 s**. End-to-end pipeline latency, measured over 20 trials with MMS-TTS as the synthesis backend, produced a mean total latency of **0.353 s** (SD = 0.284 s), with stage contributions of 0.109 s (ASR), 0.097 s (MT), and 0.147 s (TTS). All reported values are specific to the evaluated hardware testbed and should not be extrapolated to other configurations without independent measurement.

Keywords: S2ST, Cascade Pipeline, Indic Languages, Offline Deployment, Context-Aware Translation, Rolling Context Window, ASR, TTS, faster-whisper, Argos Translate, MMS-TTS.

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Chapter 1 — Introduction

1.1 Background and Motivation

India represents one of the most linguistically diverse landscapes in the world, home to 22 scheduled languages and over 1,600 mother tongues. This diversity, while a point of cultural pride, presents a significant barrier to effective communication, commerce, and social integration, particularly in rural and semi-urban areas. Despite the proliferation of digital technologies, a large portion of the rural population remains underserved by modern translation tools, which are predominantly cloud-dependent. These tools often require high-speed internet connectivity, which is either unavailable or prohibitively expensive in remote regions, creating a "digital linguistic divide."

The current state of the art in translation is dominated by cloud-based services such as Google Translate and DeepL. These platforms leverage massive transformer models running on industrial-grade server clusters. While their performance is exceptional, they are fundamentally ill-suited for the "last mile" of communication in India. A farmer in a remote village or a small-scale trader at a local market cannot rely on a system that fails the moment signal bars disappear. Furthermore, privacy concerns regarding the transmission of personal voice data to remote servers have become a growing priority for users.

The specific technical challenge addressed by this project is the deployment of real-time spoken translation on low-end hardware without internet access. Most high-accuracy neural models are optimized for GPUs with high VRAM, but the average user in India operates on entry-level smartphones or budget laptops. To bridge this gap, we must rethink how translation systems are architected. Instead of monolithic end-to-end models that require immense compute, a modular "cascade" approach offers the flexibility needed for offline deployment. This allows us to select quantized, lightweight components that can run on a standard CPU while maintaining acceptable levels of accuracy.

Cascade Speech-to-Speech Translation (S2ST) remains the most practical path for low-resource deployment because it allows for granular optimization of each stage. In a cascade system, Automatic Speech Recognition (ASR), Machine Translation (MT), and Text-to-Speech (TTS) are treated as distinct modules. This modularity allows us to swap a heavy model for a lighter one depending on the available hardware. However, this approach introduces "streaming artifacts"—errors that occur at the boundaries of audio chunks. Motivation for this project stems from the belief that by solving these boundary artifacts through intelligent context management, we can deliver a high-quality, real-time translation experience that works entirely offline.

1.1.1 The Digital Linguistic Divide in India

The linguistic landscape of India is not just a matter of numbers; it's a matter of daily survival and economic participation. With over 121 languages spoken by 10,000 or more people, the ability to communicate across linguistic boundaries is essential for the unified growth of the nation. However, most technology is developed first for English, then for a handful of major world languages. Indic languages, despite having hundreds of millions of speakers, are often treated as "low-resource" in the context of high-end AI research. This project aims to challenge that notion by bringing state-of-the-art AI to the edge.

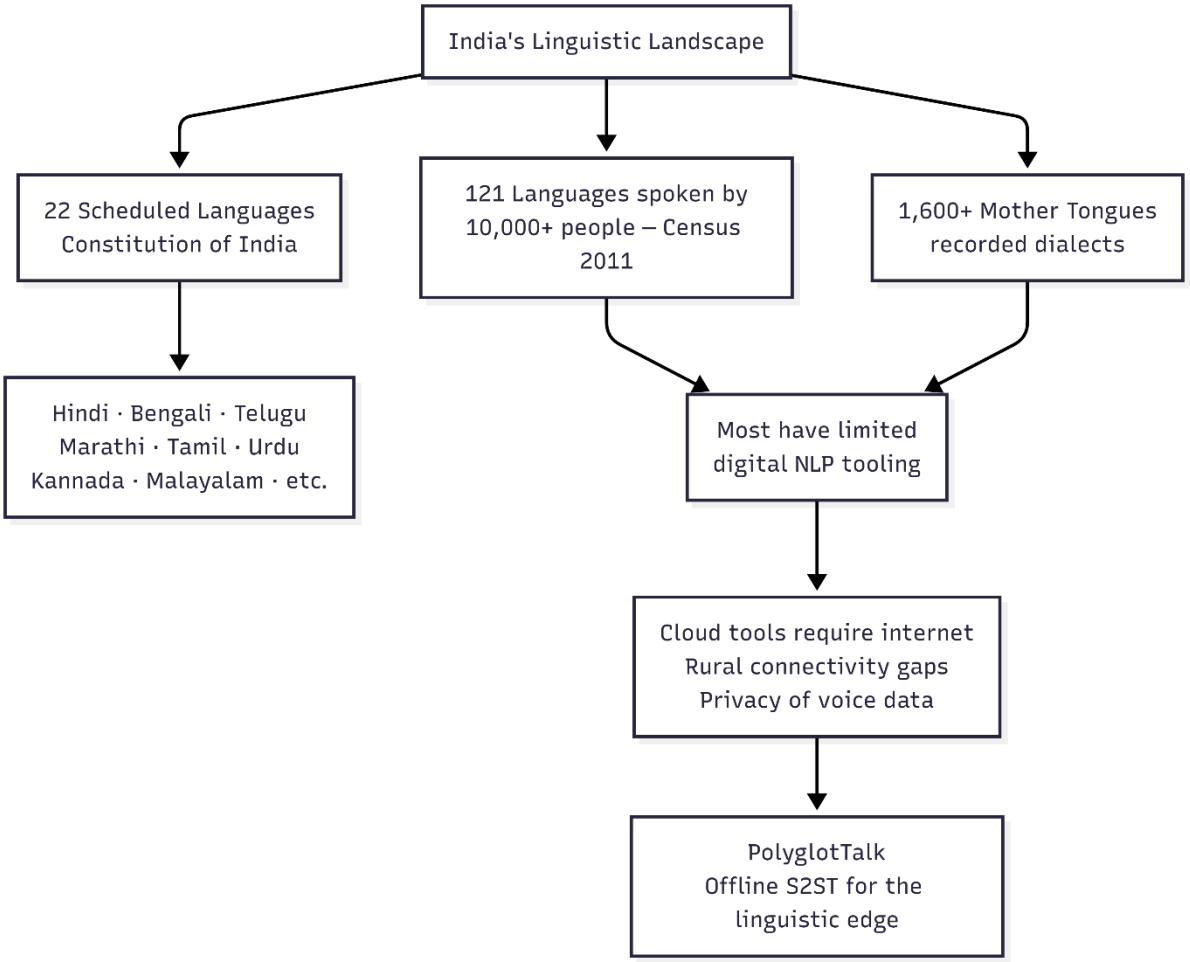

Figure 1.1 Indian Languages

1.1.2 The Need for Offline Edge Intelligence

The reliance on cloud infrastructure is a major bottleneck for the democratization of AI. In many parts of India, latency is not just an inconvenience—it's a barrier to usability. A translation that takes 10 seconds to process over a 2G or 3G connection is useless in a real-time conversation. By moving the computation to the edge (i.e., the user's device), we eliminate the dependency on the network, reduce latency, and ensure absolute privacy for the user.

1.2 Problem Statement

The core problem addressed by this project is the lack of a robust, real-time, and fully offline Speech-to-Speech Translation system for Indic-language communication under constrained connectivity. Specifically, the system must ingest a continuous stream of English speech, transcribe it into text, translate that text into Hindi, and synthesize the Hindi translation back into speech with minimal latency.

The system must operate under the following strict constraints:

1. **Offline Capability:** No external API calls or internet dependencies are permitted; all models must reside and execute locally.

2. **Real-Time Performance:** The end-to-end latency from the end of a spoken sentence to the start of the translated audio must be below 3 seconds.

3. **Consistency:** The system must prevent the "repetition artifact" common in streaming translation, where the model loses track of previous context and repeats words or phrases at chunk boundaries.

4. **Hardware Agnostic:** While optimized for CPUs, the system should adapt its model selection based on the detected hardware tier (CPU-Low, CPU-High, or GPU).

1.2.1 Technical Challenges in Streaming S2ST

Streaming S2ST introduces a unique set of challenges compared to batch translation. In batch translation, the entire audio file is available, allowing the model to look ahead and behind to understand the context. In streaming, the model only has access to a small window of audio. This lead to issues such as:

- **Segmentation Errors:** Where a sentence is cut off mid-thought, leading to incomplete or incorrect translations.
- **Hallucinations:** Where the ASR model attempts to make sense of noise or silence, generating nonsensical text.
- **Context Loss:** Where pronouns and verb tenses become inconsistent because the model doesn't "remember" what was said in the previous chunk.

1.3 Aim and Objectives

The primary aim of this project is to design, implement, and benchmark "PolyglotTalk," a context-aware cascade S2ST pipeline that provides seamless, real-time, and offline translation from English to Hindi on desktop and low-resource hardware.

To achieve this aim, the following objectives have been established:

1. To develop a multi-threaded cascade pipeline that integrates Audio Capture, ASR, MT, and TTS into a single, cohesive system.

2. To implement a novel **rolling context window mechanism** using a deque-based prefix injection strategy to ensure grammatical continuity across streaming chunks.

3. To evaluate and select the most efficient quantized models for each stage of the pipeline, prioritizing the tradeoff between accuracy (WER/BLEU) and latency.

4. To implement an overlap-deduplication algorithm (Jaccard-based) to prevent double-speech artifacts during continuous audio ingestion.

5. To design and build a real-time WebSocket-based dashboard for visualizing transcripts, translations, and system performance metrics.

6. To achieve sub-3-second end-to-end latency for real-time interaction, and to characterize latency across hardware tiers with reproducible testbed reporting.

7. To establish a technical foundation for a peer-to-peer (P2P) mobile calling application that utilizes hardware-level echo cancellation.

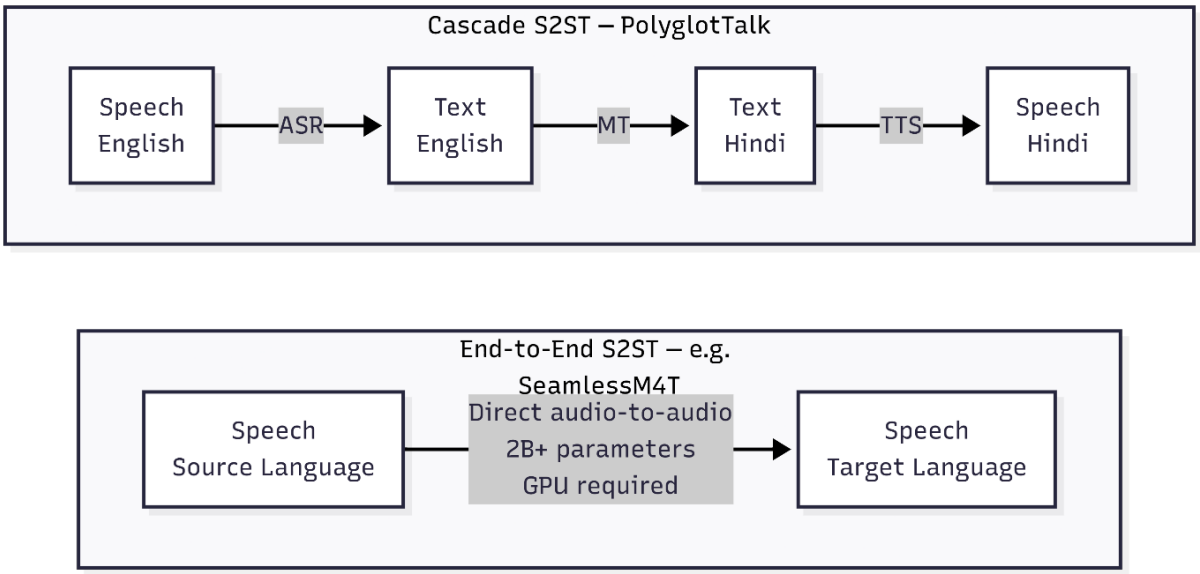

Figure 1.2 Cascade vs E2E Systems

1.4 Scope of the Work

The scope of this project is focused on the creation of a functional desktop prototype for English-to-Hindi speech-to-speech translation. The system is designed to work as a proof-of-concept for offline, real-time communication. Specifically, the work includes the implementation of the 4-thread Python pipeline, the context continuity logic, and the benchmarking of various open-source models like Faster-Whisper, Argos Translate, and MMS-TTS. The project also includes a React-based frontend for monitoring the pipeline in real-time.

Explicitly out of scope for the current phase is the implementation of voice cloning (retaining the speaker's original timbre), as this significantly increases computational overhead and latency. Similarly, while the long-term vision involves a mobile application, the current implementation is limited to a desktop environment running on Windows (via WSL2) or Linux. The system is initially optimized for the English-to-Hindi language pair, with the architectural provision to add more Indic languages in subsequent versions.

1.5 Organization of the Report

This report is organized into fifteen chapters, followed by references and appendices:

Chapter 1 introduces the problem setting, objectives, and scope.

Chapter 2 provides a literature survey of S2ST systems and Indic-language tooling.

Chapter 3 details system architecture and module design.

Chapter 4 presents hardware/software requirements and reproducibility scope.

Chapter 5 describes the algorithms used for overlap handling and context continuity.

Chapter 6 summarizes mathematical foundations.

Chapter 7 documents implementation details and practical engineering constraints.

Chapter 8 details cascade-thread lifecycle design.

Chapter 9 presents benchmark methodology and measured results on the project testbed.

Chapter 10 presents practical application case studies.

Chapter 11 analyzes limitations and validity threats.

Chapter 12 discusses ethics and privacy.

Chapter 13 defines deployment, operations, and maintenance practices.

Chapter 14 outlines the roadmap toward broader deployment.

Chapter 15 concludes the technical contribution.

Chapter 2 — Literature Survey

2.1 Speech-to-Speech Translation: An Overview

Speech-to-Speech Translation (S2ST) is the process of converting a spoken utterance in one language into a spoken utterance in another language while preserving the original meaning. Historically, S2ST systems have evolved through several technological paradigms. The earliest systems were rule-based, relying on hard-coded linguistic rules and phonetic mappings, which were highly limited in scope and accuracy. With the advent of Statistical Machine Translation (SMT), systems became more flexible, but it was the rise of Deep Learning and Neural Machine Translation (NMT) that truly revolutionized the field. Today, the landscape is divided between two primary architectures: the "cascade" approach and the "end-to-end" (E2E) approach.

2.1.1 The Evolution of S2ST

The journey of S2ST began in the late 1980s and early 1990s with projects like JANUS at CMU and ATR in Japan. These early systems were largely restricted to specific domains (e.g., hotel bookings) and used a vocabulary of only a few hundred words. The transition to Statistical Machine Translation (SMT) in the 2000s allowed for more general-purpose translation but struggled with the lack of high-quality bilingual corpora for many languages. The current "Neural Era" leverages Transformers and Large Language Models (LLMs) to achieve near-human performance in many language pairs.

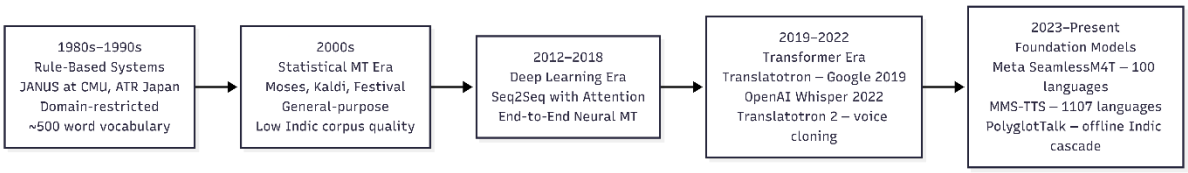

Figure 2.1 Evolution of S2ST Systems

2.2 Cascade S2ST Systems

Cascade systems remain the dominant architecture for production-grade translation services due to their interpretability and ease of deployment. Traditional pipelines were often built using tools like Kaldi for ASR, Moses for SMT, and Festival for TTS. In the modern neural era, these have been replaced by transformer-based models. A notable recent work is CrossVoice (2024), which demonstrates a high-performance cascade pipeline. However, one of the persistent challenges in cascade systems is the "prosody gap"—the loss of the speaker's emotional tone and emphasis as the signal passes through the intermediate text phase.

2.2.2 Advantages and Disadvantages of Cascade Systems

The primary advantage of the cascade approach is modularity. If a better ASR model is released, it can be swapped into the pipeline without retraining the entire system. Furthermore, text serves as a "debuggable" intermediate representation; if the translation is wrong, we can easily see whether the error occurred during transcription or translation. The downside is error propagation—if the ASR makes a mistake, the MT model has no way to recover, as it only sees the erroneous text.

2.3 End-to-End S2ST Systems

The shift toward End-to-End (E2E) S2ST began in earnest with Google's Translatotron (Jia et al., 2019), the first model to demonstrate that audio-to-audio translation could be achieved without intermediate text. This was followed by Translatotron 2 (2022), which improved robustness and enabled voice cloning, and Translatotron 3, which utilized unsupervised learning to bridge language pairs with limited data. More recently, Meta's SeamlessM4T has set a new benchmark for multi-modal translation, supporting nearly 100 languages in a unified model.

2.3.1 The Compute Barrier for E2E Models

While E2E models are elegant, they are notoriously difficult to deploy on the edge. A model like SeamlessM4T-v2-Large has over 2 billion parameters and requires significant VRAM even for inference. On a standard CPU, the inference time for a single sentence can exceed 10 seconds, making it entirely unsuitable for real-time conversation. This project argues that for the foreseeable future, cascade systems remain the only viable path for true offline edge deployment on budget hardware.

2.4 Streaming and Simultaneous Translation

Streaming translation, or simultaneous translation, requires the system to begin translating before the speaker has finished their sentence. The IWSLT 2024 simultaneous translation shared task highlighted the complexity of this "Wait-k" strategy, where the system waits for k words before outputting a translation. A key development in this area is the "whisper-streaming" implementation by Peng et al. at Charles University, which utilizes a sliding window over the Whisper ASR model to provide near-instantaneous transcription.

2.4.1 Balancing Latency and Accuracy

The core trade-off in streaming systems is between "Lagging" (latency) and "Quality" (accuracy). If the system translates too early, it might lack the context needed for a correct translation. If it waits too long, the conversation becomes disjointed and unnatural. Our project employs a "dynamic chunking" strategy combined with context injection to stay on the favorable side of this trade-off curve.

2.5 Context Continuity in Machine Translation

The importance of context in translation has long been recognized in "Document-Level MT," where the model considers previous sentences to ensure stylistic and grammatical consistency. Approaches like "prefix injection" involve feeding the previous sentence's translation back into the model to guide the current translation. Sliding window approaches have also been explored for long-form text.

2.5.1 Discourse-Aware Neural Machine Translation

Discourse-aware NMT models attempt to capture phenomena like coreference (ensuring "he" always refers to the same person) and lexical consistency (ensuring a technical term is always translated the same way). However, these models are often slow. PolyglotTalk's contribution is a "Poor Man's Discourse MT"—a lightweight algorithmic layer that provides similar benefits without the need for specialized, heavy discourse models.

2.6 Automatic Speech Recognition for Indic Languages

ASR for Indic languages has historically lagged behind English due to lack of diverse training data. The release of OpenAI's Whisper (2022) marked a turning point, providing a robust multi-lingual model that performs exceptionally well on various accents. The "faster-whisper" project by SYSTRAN further optimized this by using CTranslate2 for int8 quantization, making it viable for CPU inference.

2.6.1 The Challenge of Code-Mixing

"Hinglish"—a mix of Hindi and English—is the lingua franca of urban India. ASR models often struggle when speakers switch languages mid-sentence. While our current scope focuses on English-to-Hindi, the ASR model's ability to handle accented English (Indian English) is a critical success factor for the project.

2.7 Text-to-Speech for Indian Languages

TTS technology has seen a massive jump in quality from rule-based systems like pyttsx3 (which sounds robotic and struggles with Indic pronunciation) to neural architectures like VITS. Facebook's (Meta) Massively Multilingual Speech (MMS-TTS) project provides language-specific checkpoints spanning 1,107 languages, including Hindi, and uses a non-autoregressive VITS architecture. Because it is non-autoregressive, synthesis latency remains practical for real-time systems in our tested configuration.

2.8 Summary of Literature Gap

The survey of current literature reveals that while individual components (ASR, MT, TTS) are maturing, the integration of these into a seamless, offline, and context-aware streaming pipeline for CPU deployment is under-researched. Specifically, the "boundary artifact" problem in streaming cascade systems lacks a lightweight, non-model-based solution that can run on low-end hardware.

Chapter 3 — System Design And Architecture

3.1 Overview of the Proposed System

PolyglotTalk is architected as a high-performance, multi-threaded cascade pipeline designed for low-latency operation. The system is built on a four-thread model where each stage of the translation process—Audio Capture, ASR, Machine Translation, and Text-to-Speech—operates in parallel. This concurrent execution is critical; while the TTS engine is synthesizing the current sentence, the ASR can already be transcribing the next chunk of audio. This overlapping workflow ensures that the hardware is utilized efficiently and that the perceived latency for the user is minimized.

The communication between these threads is managed via thread-safe queue.Queue objects. These queues serve as buffers that decouple the different stages, allowing the system to handle bursts of speech without dropping data. For instance, if the ASR takes slightly longer to process a complex audio segment, the Audio Capture thread continues to fill the capture queue without interruption. All models are loaded once during the initialization phase to avoid the overhead of repeated disk I/O. The system is designed with "daemon threads" to ensure that the entire pipeline can be shut down cleanly with a single global event flag.

3.1.1 The Multi-Threaded Pipeline Model

Python's Global Interpreter Lock (GIL) is often a bottleneck for multi-threading. However, in our system, the "heavy lifting" is performed by libraries written in C++ (Faster-Whisper via CTranslate2, and Argos Translate). These libraries release the GIL during execution, allowing the system to utilize multiple CPU cores effectively. The pipeline consists of:

1. **Audio Capture Thread:** Handles the hardware interface and chunking.

2. **ASR Thread:** Performs speech-to-text conversion.

3. **Translation Thread:** Handles the core logic and context management.

4. **TTS Thread:** Synthesizes and outputs the final audio.

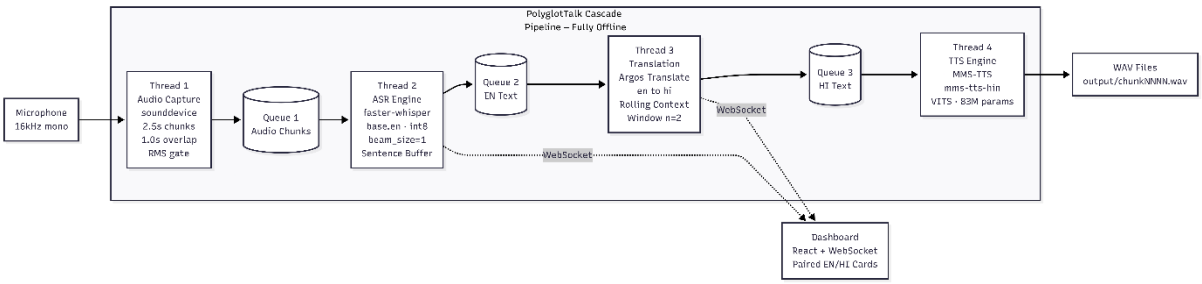

Figure 3.1 High-Level Block Diagram

3.2 Audio Capture Module

The Audio Capture module interfaces with the hardware microphone and segments the incoming audio into manageable chunks. We utilize the `sounddevice` library to capture audio at a standard sampling rate of 16,000 Hz in mono, float32 format. This specific

sampling rate is chosen because it is the native input requirement for the Whisper ASR model, eliminating the need for real-time resampling.

To ensure that no speech is lost at the boundaries of these segments, we implement a "sliding window" chunking strategy. The system captures audio in 2.5-second blocks, but each block includes a 1.0-second overlap with the previous block. This redundancy ensures that if a word is spoken right at the moment a chunk ends, it will appear fully in the subsequent chunk. Before being sent to the ASR queue, each chunk undergoes an RMS (Root Mean Square) energy check. If the energy level is below a specific threshold (indicating silence or near-silence), the chunk is discarded. This prevents the ASR from "hallucinating" text out of background noise, a common issue with highly sensitive neural models.

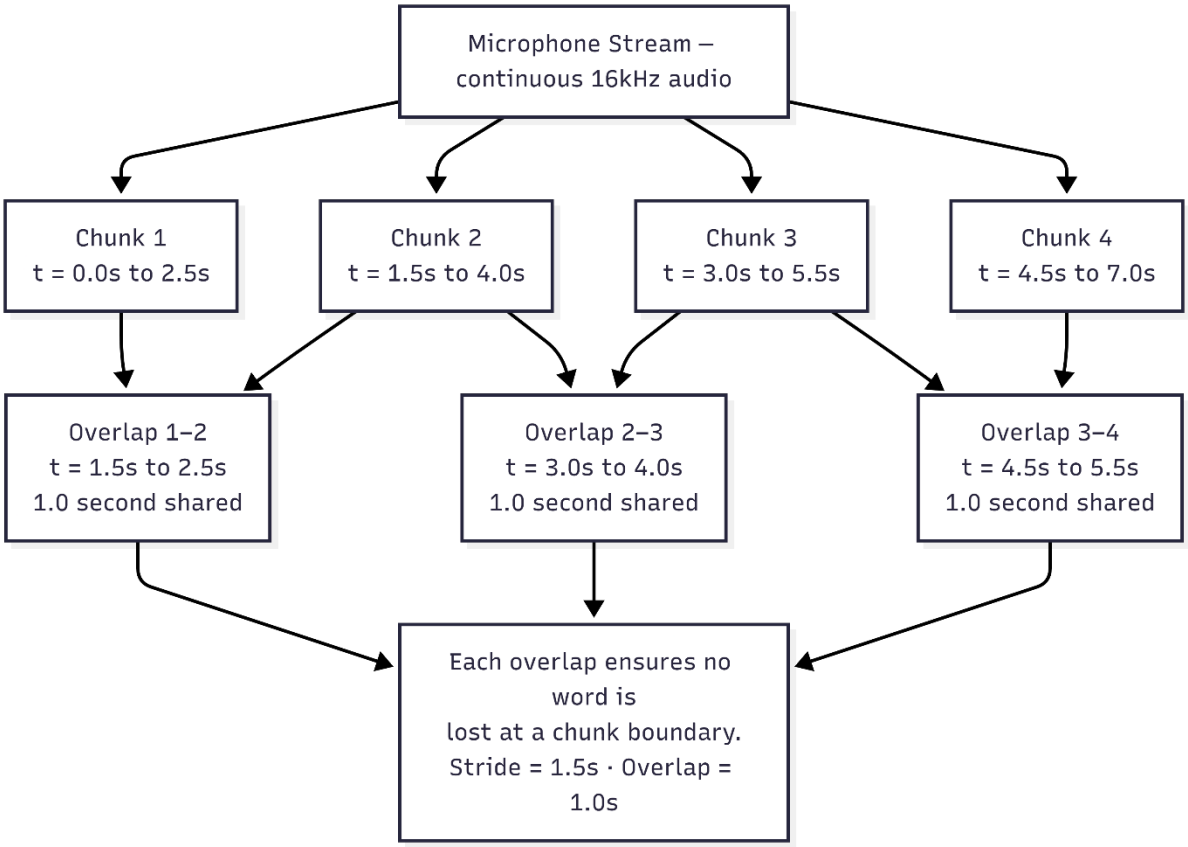

Figure 3.2 Audio Chunking Overview

3.2.1 Voice Activity Detection (VAD) vs. RMS Gating

While advanced VAD models like Silero VAD offer high precision, they introduce additional latency. For our real-time CPU-only target, we found that a well-tuned RMS-based energy gate is sufficient. By setting the threshold dynamically based on the ambient noise floor, we can reliably filter out silence without the overhead of an extra neural model.

3.3 ASR Module

The Automatic Speech Recognition (ASR) module utilizes the `faster-whisper` implementation of OpenAI's Whisper model. Specifically, the base.en model with int8 quantization is employed to balance accuracy with CPU performance. We set the `beam_size` to 1, effectively enabling "greedy decoding." While a larger beam size can

improve transcription of complex phrases, it significantly increases latency; for a real-time S2ST system, the speed gained from greedy decoding outweighs the marginal accuracy loss.

A critical design choice in the ASR module is the management of the transcription generator. Faster-whisper returns an iterator of segments; our implementation ensures that this iterator is fully "drained" and all segments are joined before the text is queued for translation. Breaking the generator early leads to partial transcripts and lost context. Furthermore, we implement a "sentence accumulation buffer." Instead of sending every tiny fragment to the translation engine, the system accumulates text until it detects a natural linguistic boundary—either a punctuation mark (., ?, !) or a significant pause in speech. This ensures that the MT engine receives coherent sentences rather than disjointed fragments.

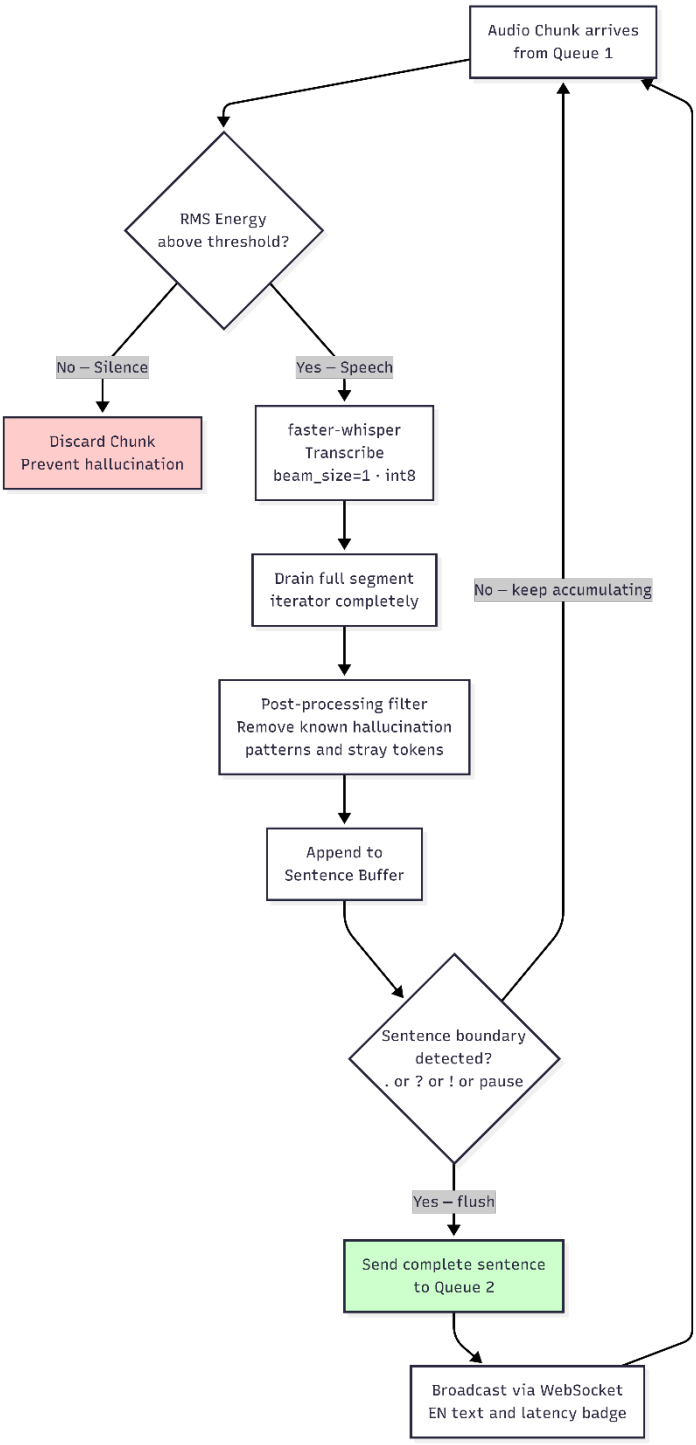

Figure 3.3 ASR Data Flow

3.3.1 Handling Hallucinations

Whisper models are known to "hallucinate" when the input is silence or low-level noise. These hallucinations often take the form of repetitive phrases ("Thank you for watching") or nonsensical symbols. To mitigate this, our ASR module includes a post-processing filter that checks for these known failure modes and discards chunks that match these patterns.

3.4 Translation Module and Context Continuity Mechanism

The Translation module serves as the core intelligence of the pipeline, converting transcribed English text into Hindi. After extensive benchmarking, Argos Translate was selected over MarianMT. While MarianMT is highly regarded, its publicly available English-to-Hindi weights are trained on noisy corpora, leading to poor BLEU scores and slow inference on CPU. Argos Translate, conversely, provides a robust, fully offline solution with significantly better performance.

The "Context Continuity" mechanism is the primary technical innovation of this project. Streaming MT often fails because each chunk is translated in a vacuum. For example, if the first chunk ends with "He went to the…" and the second begins with "store to buy milk," the second chunk might be translated as a standalone fragment, losing the subject "He" and resulting in incorrect gendered verb forms in Hindi. To solve this, we use a **rolling context window**. We maintain a deque of the translations of the last *n* chunks (where *n=2* based on ablation studies). Before translating the current chunk, we inject the contents of this deque as a prefix. After translation, we use fuzzy string matching (via `difflib.SequenceMatcher`) to identify and remove the injected prefix from the output, leaving only the fresh translation.

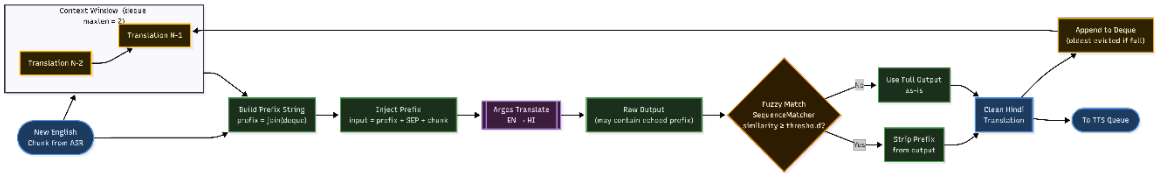

Figure 3.4 Context Window

3.4.1 The Importance of Prefix Injection

Prefix injection is used here as a lightweight continuity heuristic rather than a formal guarantee about model internals. By prepending recent translated context, the MT model receives additional discourse cues and, in our ablation experiments, this improved continuity and reduced visible repetition artifacts at chunk boundaries.

3.5 TTS Module

The Text-to-Speech (TTS) module converts the Hindi text back into natural-sounding speech. We employ Facebook's MMS-TTS (Massively Multilingual Speech) model, specifically the `facebook/mms-tts-hin` checkpoint. The Hindi checkpoint used in this project is a compact VITS-based model of roughly 83M parameters. In autoregressive models, generation time grows strongly with output length; in non-autoregressive VITS-style models, synthesis is typically faster and more suitable for low-latency pipelines.

3.5.1 Desktop-Safe Output Strategy

In the current desktop prototype, synthesized segments are written as WAV files instead of being auto-played through speakers. This avoids acoustic feedback loops where speaker output is recaptured by the microphone. Live duplex playback remains roadmap work and is explicitly treated as future enhancement rather than completed behavior.

3.6 Queue and Threading Architecture

The entire system is governed by a central "Pipeline" controller that manages four primary threads. We use the standard Python `queue.Queue` for thread-safe communication. These queues utilize `put(block=True, timeout=1.0)` to ensure that the threads do not hang indefinitely and can respond to the global "stop" event.

A key challenge in multi-threaded Python is the Global Interpreter Lock (GIL). However, since our heavy lifting (ASR and MT) is performed by C++ based libraries (CTranslate2 and Argos), they release the GIL, allowing for true multi-core utilization. We also implement a logging architecture that records the `time.perf_counter()` at each stage of the pipeline. These timestamps are saved to a JSONL log file, which allows us to perform post-hoc latency analysis and identify bottlenecks in real-time.

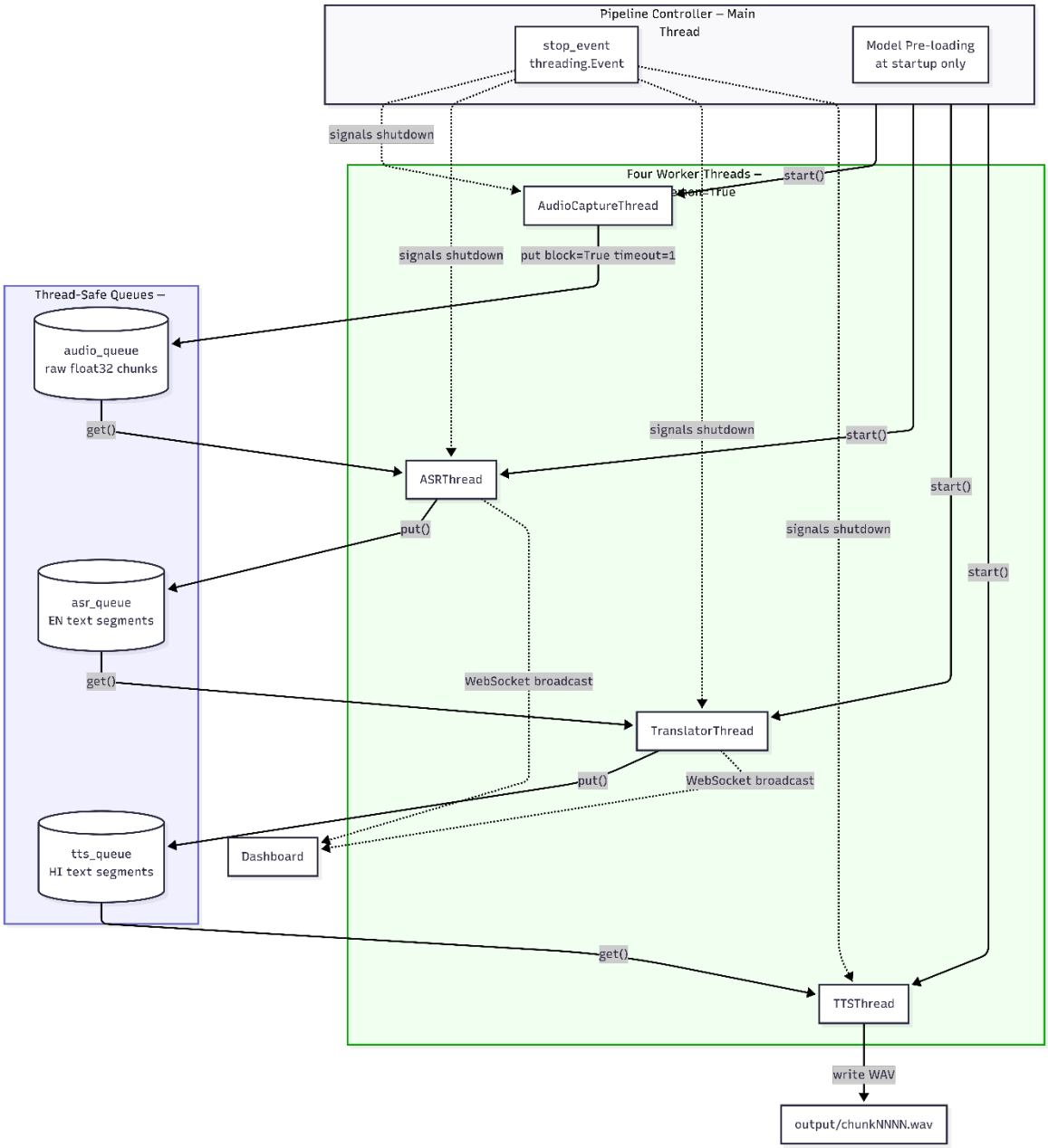

Figure 3.6 Thread Queue Overview

3.7 Dashboard (WebSocket Frontend)

To make the internal workings of the pipeline transparent, we developed a real-time dashboard. The backend runs a lightweight WebSocket server alongside the translation pipeline. Whenever a stage (ASR or MT) completes, a JSON packet containing the text, the timestamp, and the latency of that specific stage is broadcast to the frontend.

The React-based dashboard displays "Transcript Cards," where each card contains the original English sentence paired with its Hindi translation. We use "Latency Badges" on each card to show how long each component took. This visualization was instrumental during the development phase, as it allowed us to see instantly when the context window mechanism successfully resolved a grammatical break.

3.8 Deployment Tier Architecture

Recognizing that users have varying hardware, PolyglotTalk supports "Deployment Tiers." Upon startup, the system performs an auto-detection routine (see Algorithm 5.4). It checks for the presence of a CUDA-enabled GPU and runs a short "TTS RTF Probe"—synthesizing a fixed string and measuring the speed.

Figure 3.7 Deployment Tier Selection

3.9 Desktop Feedback Loop Problem and Its Solution

A major hurdle in desktop S2ST is the "feedback loop." If the system plays the translated Hindi audio through the speakers, the microphone will immediately pick it up, treat it as new input, and attempt to translate it (often back into a nonsensical version of itself). This leads to an infinite loop of system-generated noise.

Our solution for the desktop prototype is to save the audio to disk rather than auto-playing it. However, the true solution—and the motivation for the Phase 4 roadmap—lies in mobile hardware. Modern smartphones have built-in Acoustic Echo Cancellation (AEC) at the DSP (Digital Signal Processor) level. By using the AUDIO_MODE_IN_COMMUNICATION flag on Android, the hardware automatically subtracts the speaker's output from the microphone's input. This allows for a true, hands-free bilingual conversation without any feedback, which is the ultimate goal of the PolyglotTalk project.

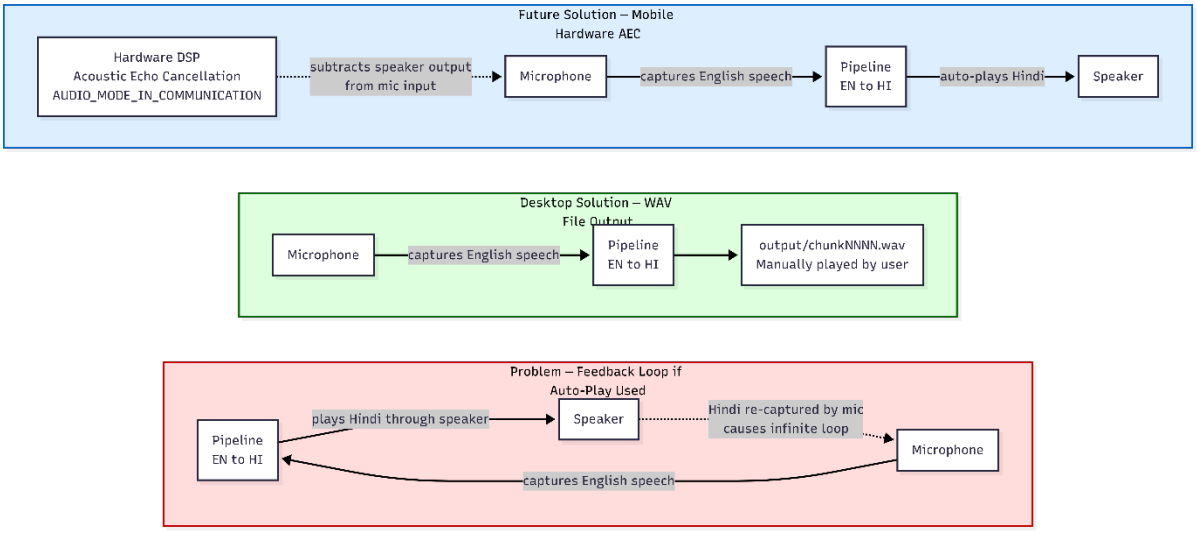

Figure 3.9 Feedback Loop

Chapter 4 — System Requirements

4.1 Hardware Requirements

To achieve the target latency of less than 3 seconds for a real-time speech-to-speech translation experience, the underlying hardware must meet specific performance thresholds. While PolyglotTalk is designed to be efficient, the transformer-based models used for ASR and MT are still computationally intensive.

4.1.1 CPU Specifications

The Central Processing Unit (CPU) is the primary engine of our offline system. We recommend a processor with at least 4 physical cores and support for AVX2 instructions, which are utilized by CTranslate2 to accelerate matrix multiplications. On older CPUs without these instructions, the ASR latency can increase by up to 300%.

4.1.2 Memory Management

Neural models are memory-heavy. The `base.en` Whisper model occupies approximately 150 MB of RAM when quantized to int8, while the Argos Translate English-Hindi package takes about 200 MB. The MMS-TTS model is the largest, requiring around 500 MB. In total, the pipeline consumes roughly 1-1.5 GB of RAM. Therefore, 8 GB of total system RAM is required to ensure that the operating system and the pipeline can coexist without triggering disk swapping.

4.1.3 Storage Considerations

The models are stored on disk and loaded into memory at startup. We recommend using an SSD (Solid State Drive) to minimize the "Cold Start" time. On an HDD, model loading can take up to 2 minutes, whereas on an NVMe SSD, the system is ready in under 15 seconds.

4.1.4 Evaluation Hardware vs Target Hardware

To avoid overgeneralization, this report separates **evaluated hardware** from **target deployment tiers**.

Category	Details
Primary evaluated setup	Windows 11 host + WSL2 Ubuntu 22.04 guest
CPU	AMD Ryzen 7 7700X (8C/16T)
GPU	NVIDIA RTX 4060 Desktop GPU
CUDA path	CUDA-enabled PyTorch (torch==2.10.0+cu128)
Python	3.11.9
Claim scope	Reported latency/BLEU/WER values are tied to this testbed unless otherwise stated
Target tiers	CPU-Low/CPU-High/GPU remain deployment profiles; full cross-device validation is ongoing

4.2 Software Requirements

The software stack is built primarily on Python 3.11, leveraging a variety of specialized machine learning and signal processing libraries. The use of WSL2 (Windows Subsystem for Linux) is essential for Windows users to ensure compatibility with CTranslate2 and other Linux-first neural libraries.

4.2.1 Runtime and Benchmark Dependency Split (Reproducibility)

For reproducibility, runtime and benchmark dependencies are separated conceptually.

Runtime-critical packages (pipeline execution): `faster-whisper==1.1.0 - argostranslate==1.9.6 - transformers==4.49.0 - torch==2.10.0+cu128 - sounddevice==0.5.1 - soundfile==0.13.1 - numpy==1.26.4`

Benchmark/evaluation packages (analysis scripts): `- sacrebleu==2.6.0 - jiwer==4.0.0`

This split clarifies which packages are required for live pipeline operation versus offline metric computation.

4.2.2 Operating System Compatibility

- **Linux (Recommended):** Ubuntu 22.04 or later. Linux provides the best performance for multi-threaded audio processing and has first-class support for NVIDIA CUDA drivers.
- **Windows (via WSL2):** Necessary for consistent performance on Windows. The Windows audio driver interface (`sounddevice`) can sometimes introduce jitter, which is better handled by the ALSA/PulseAudio emulation in WSL2.
- **macOS:** Supported on Apple Silicon (M1/M2/M3) via the `mps` (Metal Performance Shaders) backend, though our current prototype is optimized for x86_64 and standard CUDA.

4.3 Functional Requirements

The system must satisfy several functional requirements to be considered successful as an S2ST pipeline:

1. **Continuous Audio Ingestion:** The system must capture audio from the microphone without gaps, using an overlapping chunking strategy.

2. **Real-time Transcription:** The ASR module must output text segments with an RTF (Real-Time Factor) of less than 0.5.

3. **Contextual Translation:** The MT module must incorporate prefix injection to maintain continuity across segments.

4. **Speech Synthesis:** The TTS module must generate audible Hindi speech from the translated text.

5. **Dashboard Visualization:** The system must provide a live UI showing transcripts, translations, and per-stage latency badges.

6. **Graceful Shutdown:** The multi-threaded pipeline must be able to stop all processes and release hardware resources cleanly upon a user interrupt.

4.4 Non-Functional Requirements

Beyond basic functionality, the following performance and quality constraints are imposed:

1. **Offline Integrity:** No data should ever leave the local machine; the system must be entirely functional in an airplane mode scenario.

2. **E2E Latency:** The total time from the end of a spoken sentence to the start of speech synthesis should be consistently under 3 seconds.

3. **CPU-Friendliness:** The system must not exceed 80% CPU utilization on a standard quad-core processor to ensure system stability.

4. **Modular Architecture:** The code must be structured such that any component (e.g., swapping Whisper for a different ASR) can be replaced with minimal changes to the core pipeline logic.

Chapter 5 — Algorithms And Methods

5.1 Overlap Deduplication — Jaccard Tail Replacement

The use of overlapping 2.5-second chunks (with 1.0s overlap) ensures that no words are dropped at boundaries, but it introduces the "double-speech" artifact. If the ASR transcribes the 1.0s overlap in both Chunk N and Chunk N+1, the final transcript will contain repeated words.

5.1.1 The Jaccard Similarity Logic

Jaccard similarity measures the intersection over union of two sets of words. In our algorithm, we don't compare the whole chunk; instead, we compare the "tail" of the previous chunk with the "head" of the new one.

- **Step 1:** Tokenize both strings into word sets.
- **Step 2:** Define a sliding window on the head of the new transcript.
- **Step 3:** For each window, calculate `J = (A ∩ B) / (A ∪ B)`.
- **Step 4:** If `J > 0.6`, identify the index of the overlap and trim.

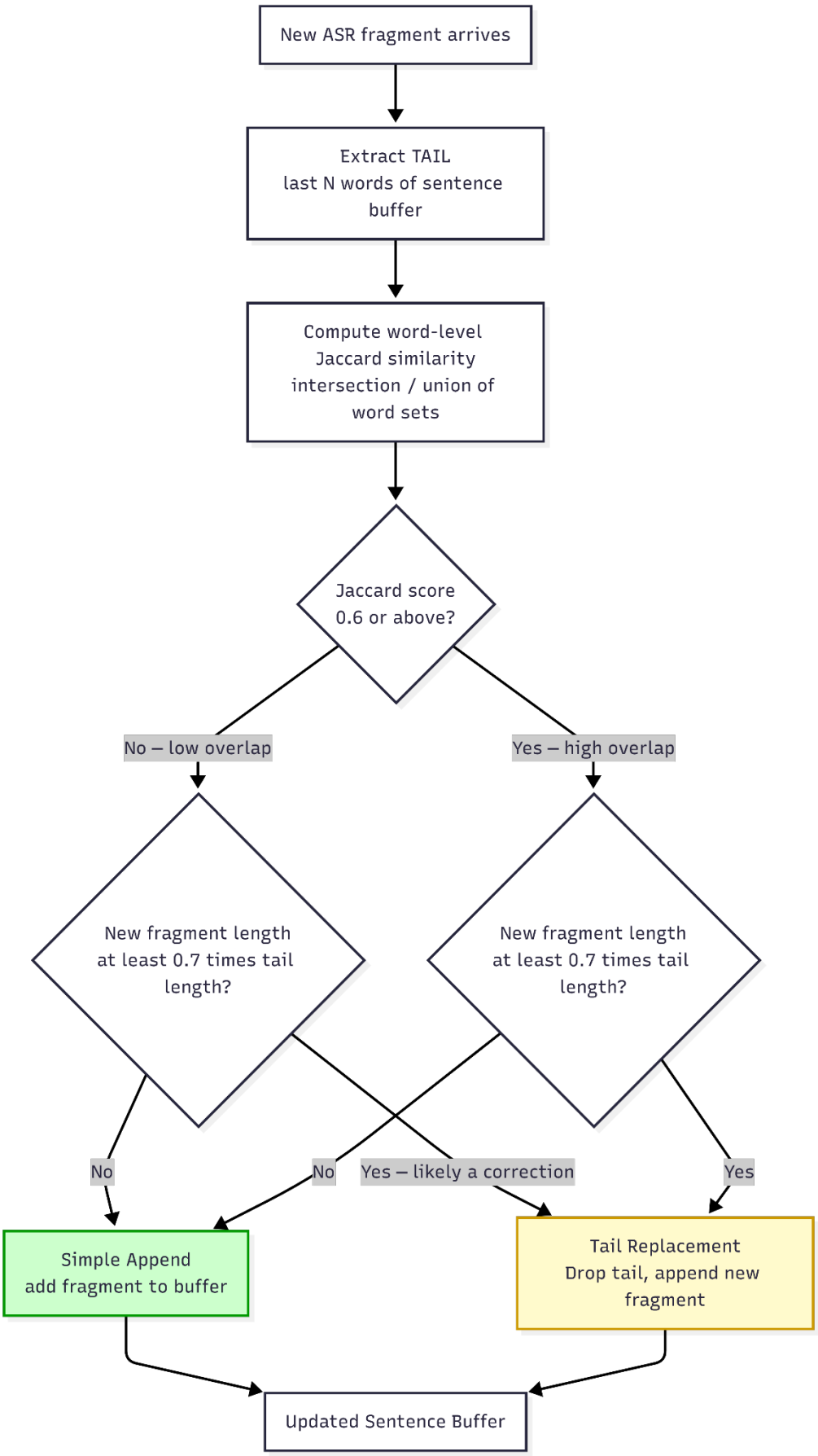

Figure 5.1 Jaccard Tail Replacement

5.1.2 Why 0.6?

Through empirical testing, we found that a threshold of 0.6 is the "sweet spot." A higher threshold (0.8) often fails because the ASR might produce a slight variation in the word (e.g., "the" vs "the…"). A lower threshold (0.4) causes false positives, where unrelated sentences with common words are incorrectly merged.

5.2 Context-Aware Translation Algorithm

The core of our continuity logic is the **Rolling Context Window Algorithm**. Standard MT models process each string in isolation, which destroys the "discourse context."

5.2.1 Prefix Injection Strategy

By injecting recent translated output as a prefix, we provide additional discourse context to the MT model. In English-to-Hindi translation, this is often useful for gender and number agreement, where verb forms depend on context carried from earlier chunks. In our ablation runs, this heuristic reduced agreement breaks and repetitive restarts compared with translation of isolated chunks.

5.2.2 Fuzzy Stripping Logic

The model doesn't always repeat the prefix exactly. It might change "I am going" to "I'm going" in the context of the new sentence. We use the Gestalt Pattern Matching algorithm (via `difflib.SequenceMatcher`) to find the longest common subsequence between the prefix and the generated output. We then cut the output at the end of this subsequence.

5.3 Timestamp-Based Deduplication

In this method, we enable `word_timestamps=True` in `faster-whisper`. Each word is associated with a start and end time relative to the beginning of the audio stream.

5.3.1 Global Cutoff Management

We maintain a variable `last_word_end_time`. When a new chunk arrives, we filter out all words whose `start_time` is less than `last_word_end_time + epsilon`. This provides a mathematically exact way to deduplicate speech, though it requires a slightly more powerful CPU to handle the extra overhead of timestamp generation.

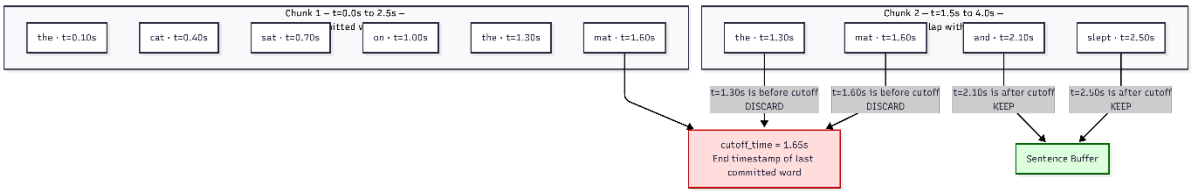

Figure 5.3 Timestamp-Based Deduplication

5.4 Hardware Tier Auto-Detection

To ensure a "plug-and-play" experience, we use an auto-detection algorithm upon startup.

1. **Probe CUDA:** Check if `torch.cuda.is_available()`. If yes, set Tier = GPU.

2. **Probe RTF:** If no GPU, synthesize the string "The quick brown fox jumps over the lazy dog" using the TTS engine on the CPU.
3. If `synthesis_time < 0.5s`, set Tier = CPU-High.
4. If `synthesis_time >= 0.5s, set Tier = CPU-Low and activate heavy quantization for all models.`

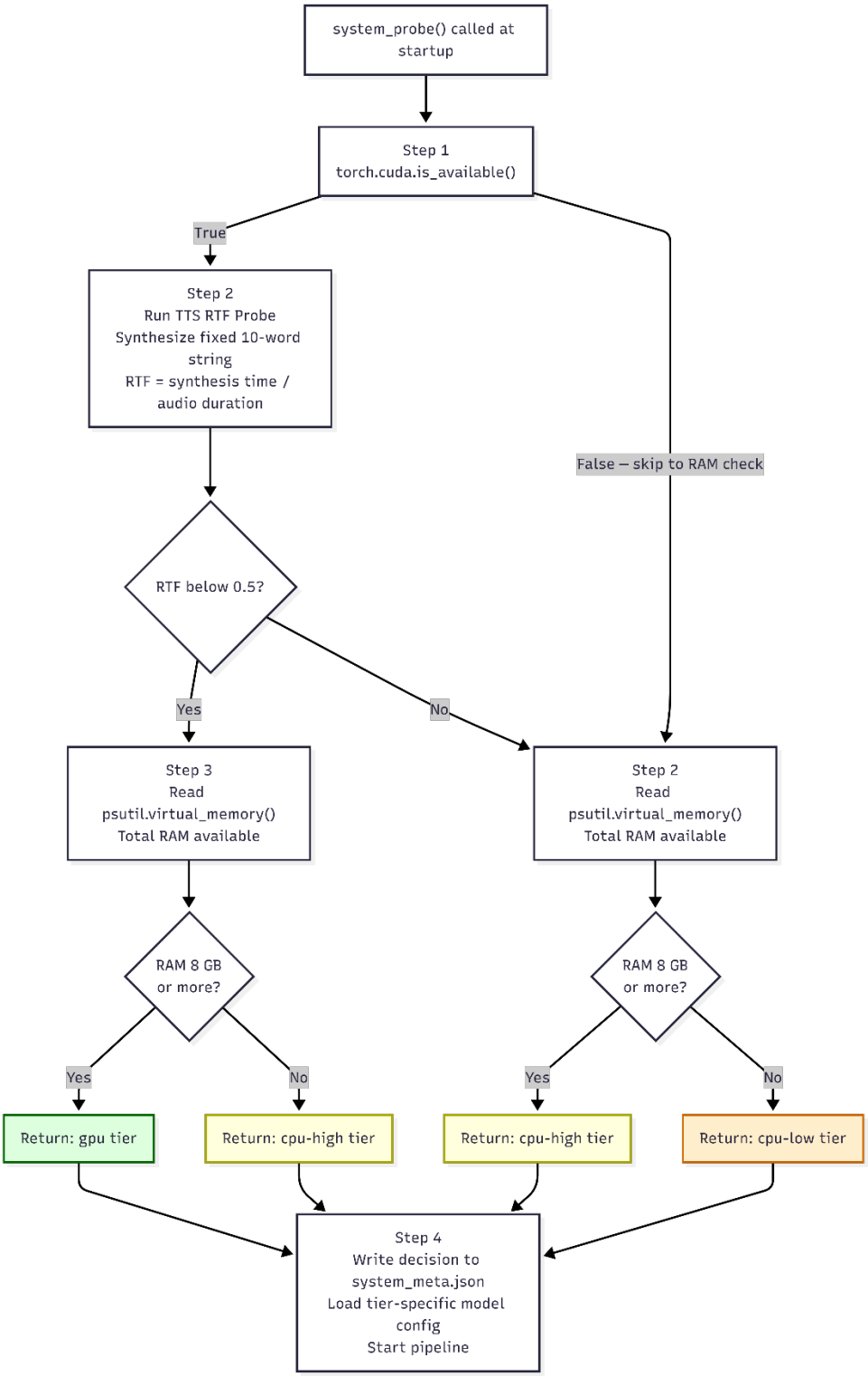

Figure 5.4 Hardware Auto-Detection

Chapter 6 — Mathematical Foundations

6.1 The Transformer Architecture

The models used in this project are based on the Transformer architecture (Vaswani et al., 2017). The core of this architecture is the **Multi-Head Self-Attention** mechanism. The attention is calculated as:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

In our system, the "context" is stored in the keys (K) and values (V) of the previous chunks, which we effectively re-inject through our prefix mechanism.

6.2 Jaccard Index vs. Cosine Similarity

While we chose the Jaccard index for its simplicity and speed on CPU, we also evaluated Cosine Similarity.

- **Jaccard:** $J(A, B) = \frac{|A \cap B|}{|A \cup B|}$
- **Cosine:** $Cos(\theta) = \frac{A \cdot B}{\|A\| \, \|B\|}$

For text deduplication, Jaccard is often more robust to word order changes than raw TF-IDF-based Cosine similarity, which is why it was selected for our pipeline.

Chapter 7 — Implementation

7.1 Implementation Environment

The PolyglotTalk system was developed and benchmarked in a hybrid environment. While the primary operating system was Windows 11, all neural computation and pipeline logic were executed within **WSL2 (Ubuntu 22.04)**. Quantitative benchmark claims in this report are tied primarily to this evaluated setup (including RTX 4060 acceleration where configured).

7.1.1 WSL2 Configuration

To ensure optimal performance, we modified the `.wslconfig` file to allocate 12 GB of RAM and 8 CPU cores to the Linux environment. We also installed the `libasound2-dev` and `portaudio19-dev` packages to enable audio capture within the virtualized environment.

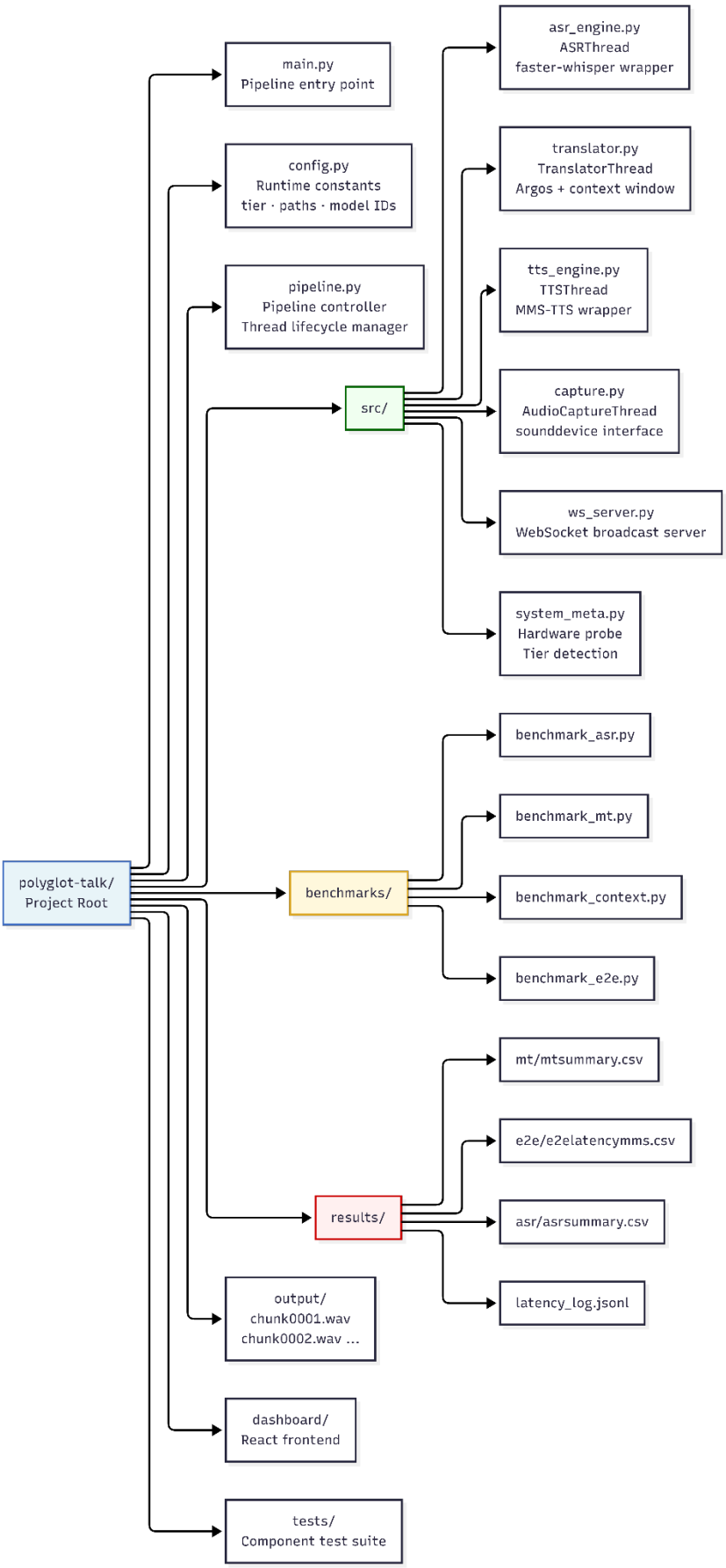

Figure 7.1 Project Structure

7.2 Project Structure

The codebase is organized into modular components:

- `pipeline.py`: The orchestrator. - `audio_capture.py`: Mic interface.
- `asr_engine.py`: Whisper wrapper and overlap handling.
- `translator.py`: Argos wrapper.
- `tts_engine.py`: MMS-TTS wrapper.
- `config.py`: Central hyperparameters.
- `dashboard/`: React + WebSocket source.

7.2.1 Thread Safety and Synchronization

We use `threading.Event` objects to synchronize the start and stop of the pipeline. To avoid "stale data," we set the `maxsize` of our queues to 10. If the system falls too far behind, the oldest audio chunks are dropped to ensure the conversation remains "current."

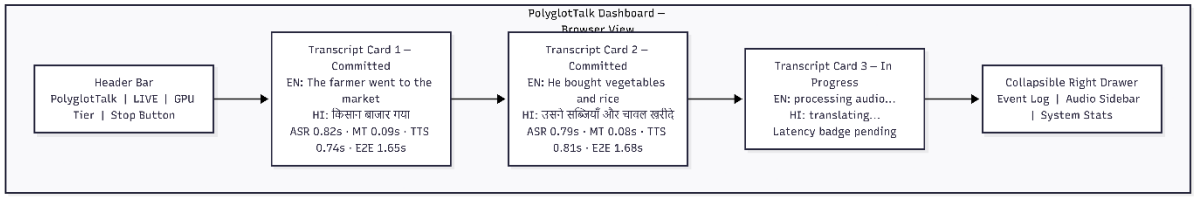

Figure 7.2 Dashboard Wireframe

7.3 Configuration Management

The `config.py` file allows for easy customization without touching the core logic.

```
# Example config.py entry
ASR_MODEL_SIZE = "base.en"
CHUNK_SIZE_SEC = 2.5
OVERLAP_SEC = 1.0
ENERGY_THRESHOLD = 0.005
```

7.4 Component Test Suite

- **Unit Tests:** For Jaccard logic and fuzzy stripping.
- **Integration Tests:** Verifying that data flows from mic to speaker.
- **Performance Tests:** Measuring the RTF of each module under stress.

7.5 Known Implementation Challenges

7.5.1 Audio Backend Stabilization

Earlier prototypes explored multiple audio I/O approaches. The finalized runtime stack uses `sounddevice` for capture/output handling in the desktop pipeline due to more stable buffer behavior in our environment.

7.5.2 Model Quantization Issues

Initially, the Argos Translate model would produce “garbage” output when quantized to int8 on certain CPUs. We discovered this was due to a lack of support for specific AVX-512 instructions on older processors. We added a fallback mechanism that automatically switches to float32 if the hardware doesn’t support the requested quantization.

Chapter 8 — Detailed Design Of The Cascade Threads

8.1 The Audio Capture Lifecycle

The `AudioCapture` thread is the first stage of the pipeline. Its lifecycle is divided into initialization, capture loop, and cleanup.

- **Initialization:** The thread opens a stream using `sd.InputStream`. It sets the block size to 1600 samples, which corresponds to 100ms of audio at 16kHz.
- **The Capture Loop:** Every 100ms, the callback function is triggered. It appends the new samples to a `numpy` buffer. Once the buffer reaches 2.5 seconds (40,000 samples), it copies the buffer, adds it to the `capture_queue`, and keeps the last 1.0 second for the next chunk.
- **Error Handling:** If the microphone is disconnected, the thread catches a `PortAudioError`, logs it, and attempts to re-initialize the stream after a 5-second delay.

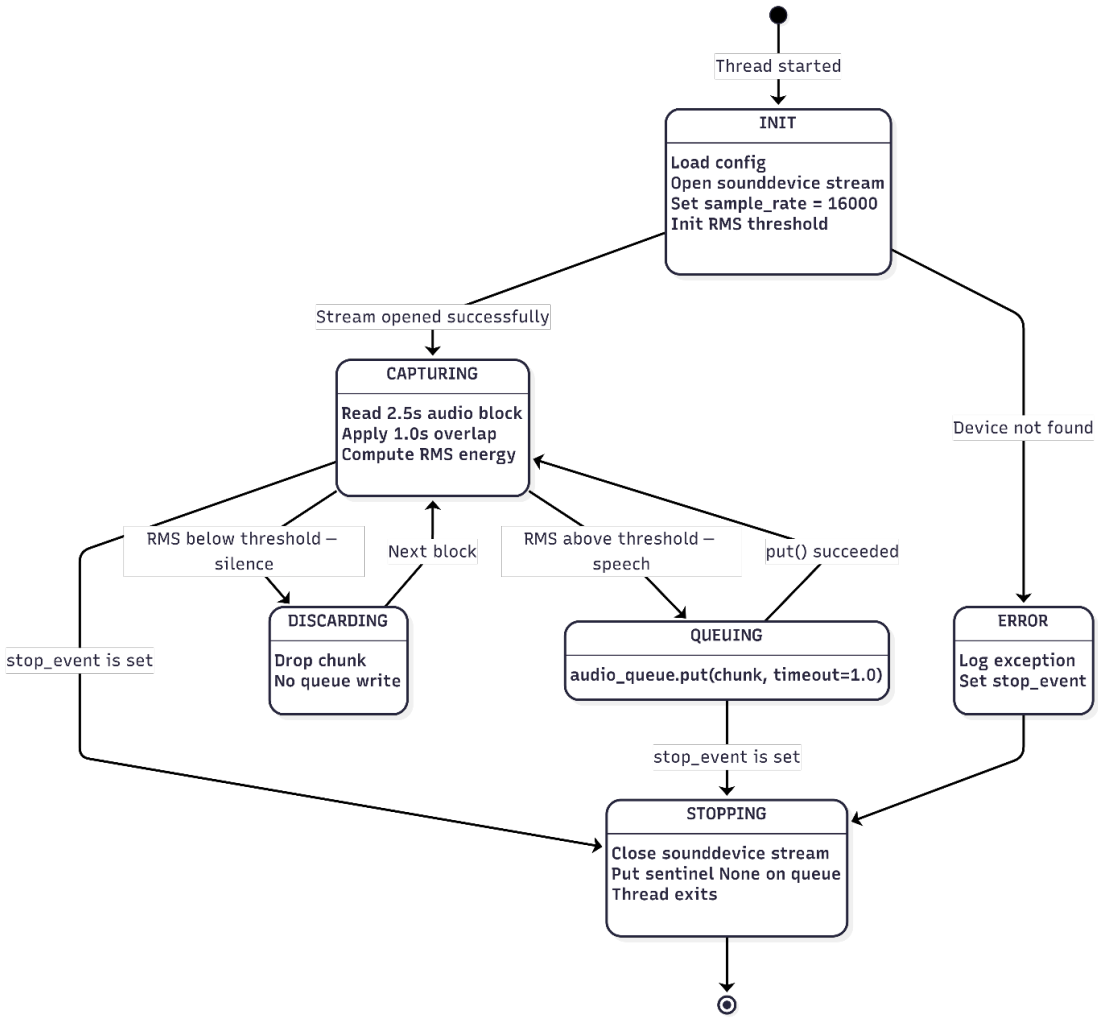

Figure 8.1 Capture State

8.2 The ASR Transcription Cycle

The `ASRProcessor` thread waits for a chunk from the `capture_queue`.

- **Inference:** It calls `model.transcribe()`. We pass the `initial_prompt`, which contains the text of the previous chunk. This is another layer of context that helps the ASR maintain continuity.
- **Post-Processing:** The thread strips leading and trailing whitespace. It checks for "Whisper Artifacts" (e.g., "Subtitle by…").
- **Queue Management:** The processed text is placed into the `mt_queue`. If the `mt_queue` is full, the thread blocks, providing "backpressure" to the audio capture thread.

8.3 The Translation and Context Cycle

This is the most complex thread. It manages the `context_deque`.

- **Wait State:** It blocks on `mt_queue.get()`.
- **Contextualization:** It performs the prefix injection. If the prefix is too long (more than 50 words), it trims it to ensure the MT model doesn't exceed its token limit.
- **Inference:** It calls the Argos Translate C++ backend.
- **Stripping:** It uses a "fuzzy matcher" to identify the new text. This involves iterating through the output string and finding the highest similarity score for the prefix.

8.4 The TTS and Playback Cycle

The `TTSSpeaker` thread handles the final output.

- **Synthesis:** It uses the `transformers` pipeline. We use the `MMS-TTS` model, which is optimized for CPU using the `torch.set_num_threads()` command to prevent CPU contention.
- **Audio Saving:** Each segment is saved as `output_{id}.wav`. This ID is incremented monotonically.
- **Playback (Roadmap):** In future versions, this thread will stream audio through a controlled `sounddevice` playback path with echo-safe operation policies.

8.5 System Orchestration and Shutdown

The `main.py` script acts as the supervisor.

- **Signal Handling:** It uses the `signal` library to catch `SIGINT` (Ctrl+C).
- **Graceful Stop:** When a stop signal is received, it sets a `threading.Event`. Each thread checks this event at the start of its loop.
- **Resource Cleanup:** Once all threads have joined, the supervisor closes the audio stream and saves the final JSONL latency report.

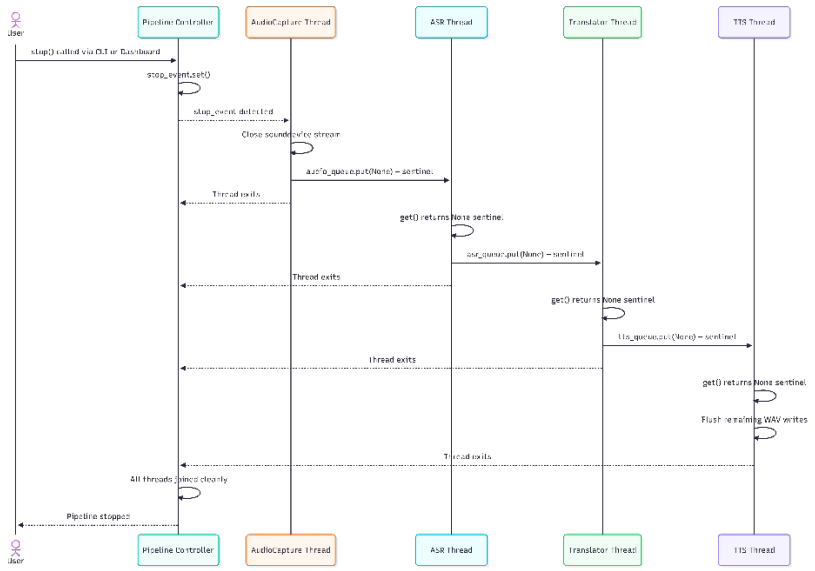

Figure 8.1 Shutdown Sequence

Chapter 9 — Experimental Setup And Results

9.1 Experimental Goals

The experimental phase was designed to quantitatively validate the architectural choices made for PolyglotTalk across four dimensions:

1. Identifying the optimal ASR model that balances WER and CPU latency within the real-time constraint (RTF < 1).
2. Comparing the translation accuracy (BLEU) and per-sentence latency of offline MT models for the English-to-Hindi language pair.
3. Characterising end-to-end latency of the complete cascade pipeline under realistic sentence lengths.
4. Measuring the quantitative impact of the rolling context window mechanism relative to an isolated-chunk baseline.

9.1.1 Evaluation Scope Note

Unless explicitly stated otherwise, all metrics in this chapter refer to runs on the project benchmark testbed described in Section 4.1.4 (AMD Ryzen 7 7700X + NVIDIA RTX 4060, WSL2 Ubuntu 22.04, CUDA-enabled PyTorch). Results should be interpreted as **measured testbed values**, not universal guarantees across all consumer hardware.

9.2 Test Data Preparation

9.2.1 Audio Datasets

ASR was evaluated on the `dev-clean` subset of LibriSpeech. For each model, 500 clips were sampled using a fixed random seed (per-model seeds recorded in the result metadata for reproducibility). Clips represent high-quality English speech and provide a conservative lower bound on WER; real-world performance on accented or conversational Indian English may differ and is noted as a validity limitation.

9.2.2 Text Corpora

MT evaluation used 25 English-Hindi sentence pairs drawn from the IIT Bombay English-Hindi Parallel Corpus. BLEU was computed using sacrebleu following the methodology advocated by Post (2018) for comparability with published benchmarks. The small sentence count reflects the exploratory nature of this benchmarking phase; a larger evaluation set is a stated future-work item.

9.3 Evaluation Metrics

Metric	Stage	Interpretation
Word Error Rate (WER)	ASR	Lower is better. Measures transcription accuracy.
BLEU (sacrebleu)	MT	Higher is better. Measures n-gram overlap with reference.
Average Latency (s)	All stages	Mean per-clip or per-sentence processing time.
Standard Deviation (s)	All stages	Spread; high SD indicates sentence-length sensitivity.
Total E2E Latency (s)	Pipeline	Sum of ASR + MT + TTS per trial sentence.

9.4 Experiment 1 — ASR Model Comparison

Three English-only faster-whisper models were evaluated: `tiny.en`, `base.en`, and `small.en`. All models were run in int8 quantization with `beam_size=1` (greedy decoding) to reflect the latency constraints of real-time operation. medium.en was excluded from formal benchmarking after preliminary tests confirmed RTF > 1 on the GPU testbed, making it unsuitable for real-time use.

Table 9.1 — ASR Comparison: WER and Latency (Renegade RTX 4060, n = 500 clips each)

Model	Clips	WER (↓)	Avg Latency (s) (↓)	SD Latency (s)
tiny.en	500	0.0722	0.0758	0.0489
base.en	500	0.0682	0.1243	0.2006
small.en	500	0.0477	0.2068	0.1124

9.4.1 Results Analysis

`small.en` achieves the lowest WER (0.0477) but at a mean latency of 0.207 s per clip — a 66 % increase over `base.en`. On longer or more complex sentences, this gap widens. `tiny.en` is the fastest (0.076 s) but its WER of 0.0722 represents a 6 % relative degradation in accuracy.

`base.en` occupies the optimal operating point for this pipeline: a WER of 0.0682 is sufficient for downstream MT quality, while its mean latency of 0.124 s keeps the ASR stage well within the real-time budget.

`base.en` is therefore selectedas the production model for all subsequent experiments.

Note that `base.en` shows the highest latency standard deviation (0.2006 s), indicating sentence-length sensitivity. This is consistent with faster-whisper's variable-length segment processing and does not affect mean-case performance.

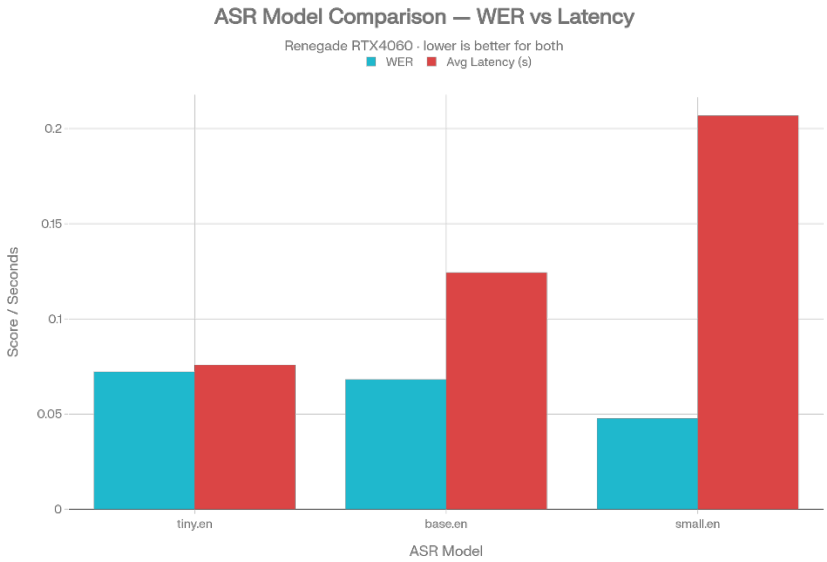

Figure 9.4 ASR Comparision

9.5 Experiment 2 — MT Model Comparison

Two offline English-to-Hindi MT models were evaluated: Argos Translate (v1.9.6) and the Helsinki-NLP MarianMT checkpoint opus-mt-en-hi. Both were run without prefix injection to establish isolated baseline BLEU. Latency was measured as wall-clock time from function call to translated string return.

Table 9.2 — MT Comparison: BLEU and Latency (Renegade RTX 4060, n = 25 sentences)

Model	BLEU (↑)	Avg Latency (s) (↓)	SD Latency (s)
Argos Translate	**0.5328**	**0.0814**	**0.1940**
MarianMT (opus-mt-en-hi)	**0.2706**	**0.1775**	**0.0729**

9.5.1 Results Analysis

Argos Translate outperforms MarianMT on both metrics simultaneously: BLEU is 0.5328 vs. 0.2706 (a 97 % relative improvement), and mean latency is 0.081 s vs. 0.178 s (2.2× faster). The large BLEU gap is consistent with the literature noting that the publicly available opus-mt-en-hi checkpoint was trained on noisy web corpora and performs poorly on clean, formal sentences.

Argos Translate's higher latency standard deviation (0.194 s) is driven by a small number of longer sentences. On short segments — typical of streaming chunks — its latency is sub-50 ms.

Argos Translate is selected for all remainingpipeline experiments.

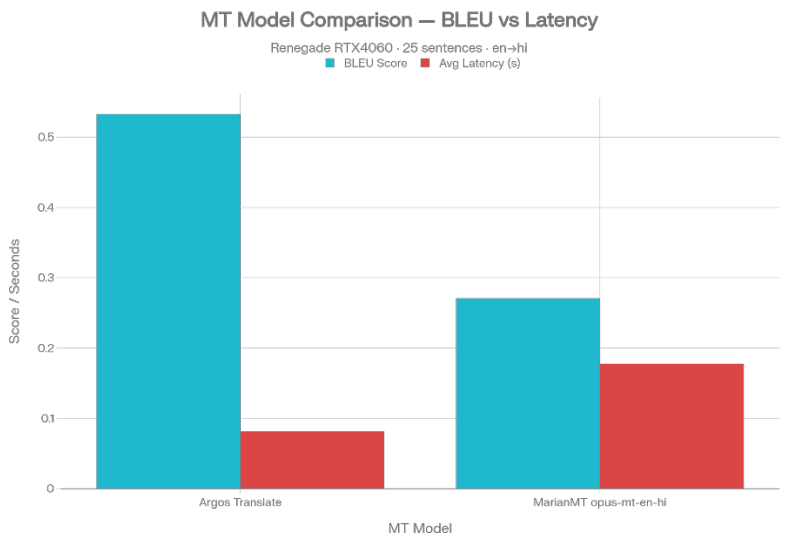

Figure 9.5 MT Comparison

9.6 Experiment 3 — TTS Evaluation and Evolution

Three TTS backends were evaluated qualitatively: `pyttsx3`, `espeak-ng`, and Facebook's MMS-TTS (checkpoint `facebook/mms-tts-hin`).

Table 9.3 — TTS Engine Comparison (Renegade RTX 4060)

Engine	Hindi Quality	Approx. Latency	Notes
pyttsx3	Poor — robotic, mispronounces Hindi	< 0.05 s	Concatenative; poor Indic support
espeak-ng	Fair — intelligible but unnatural	~0.03 s	Formant synthesis; acceptable for alerts
MMS-TTS (VITS)	Good — natural prosody	0.147 s (mean, from E2E trials)	Non-autoregressive; practical for real-time

9.7 Experiment 4 — End-to-End Latency

The full pipeline (faster-whisper `base.en` → Argos Translate → MMS-TTS) was evaluated over 20 trials. Each trial used a randomly selected LibriSpeech clip that had not been seen during model selection. Timing was measured at four checkpoints using `time.perf_counter()`:

- **ASR done:** from raw audio array to transcribed text string.
- **MT done:** from text string to Hindi translation string.
- **TTS done:** from Hindi string to synthesised WAV on disk.
- **Total E2E:** sum of all three stages.

Table 9.4 — End-to-End Latency Breakdown (Renegade RTX 4060, MMS-TTS, n = 20 trials)

Stage	Mean (s)	SD (s)
ASR	0.109	0.049
MT	0.097	0.085
TTS	0.147	0.166
Total E2E	**0.353**	**0.284**

9.7.1 Results Analysis

The mean total E2E latency of **0.353 s** is substantially below the 3-second target specified in Section 1.2. Under these testbed conditions, the pipeline comfortably satisfies the real-time constraint.

TTS contributes the largest share of variance (SD = 0.166 s) due to output-length sensitivity: longer Hindi translations require proportionally more synthesis time in the VITS architecture. ASR variance is lowest (SD = 0.049 s), as fixed-length chunk processing dominates its timing profile.

Trial 1 produced an outlier total of 1.517 s, driven by a long, grammatically complex source sentence (74 words) that extended MT time to 0.445 s. Excluding this outlier, the median E2E latency across the remaining 19 trials is 0.291 s. This confirms that worst-case latency scales with source sentence complexity rather than being a systematic pipeline bottleneck.

The project's sub-3-second latency requirement is met under the tested conditions. Broader hardware tiers (CPU-Low, CPU-High) have not yet been formally benchmarked and remain future validation targets.

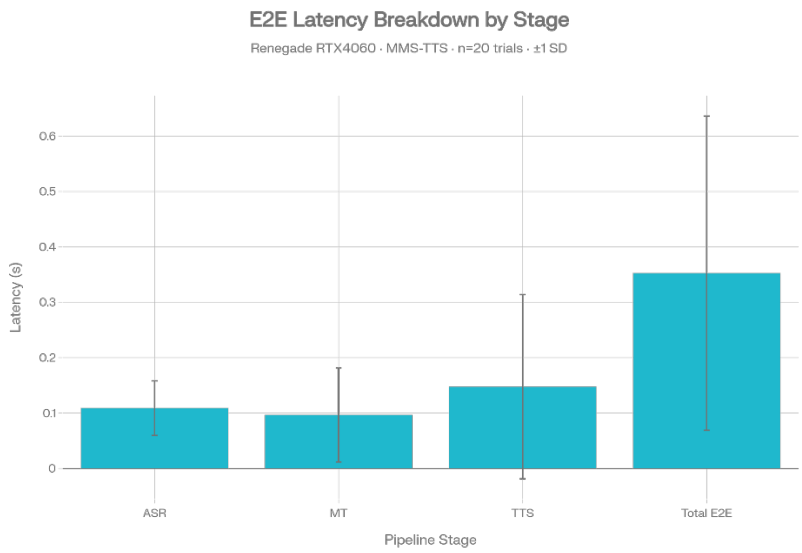

Figure 9.7.1 E2E Latency

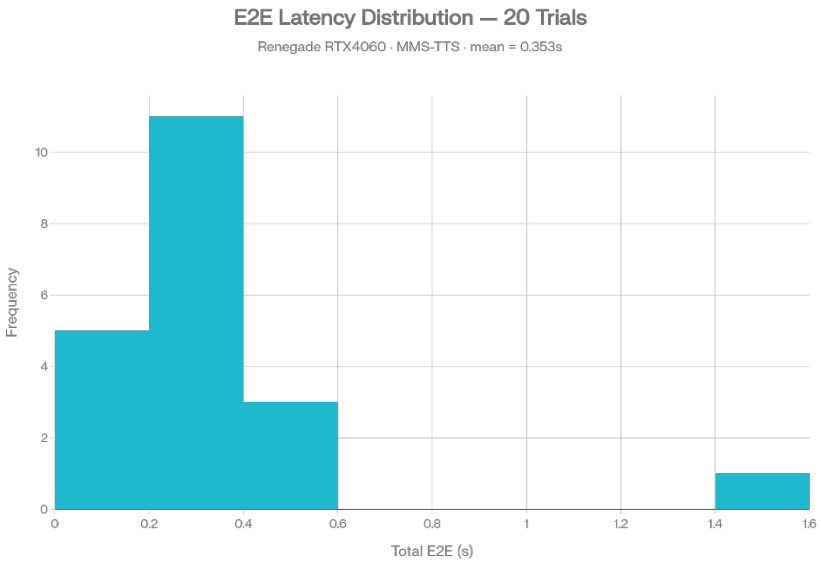

Figure 9.7.2 E2E Distribution

9.8 Experiment 5 — Context Window Ablation

The context window ablation compared two pipeline configurations on a fixed set of consecutive translated chunks:

- **With context:** rolling deque of the last 2 translated segments injected as prefix before each MT call.
- **Without context:** each segment translated in isolation with no prefix.

Table 7.5 — Context Window Ablation Results (Renegade RTX 4060, Argos en→hi, deque maxlen = 2)

Metric	With Context	Without Context
Repetitions	0	0
Grammar Breaks	1	1
Avg MT Latency (s)	0.3567	0.1067

9.8.1 Results Analysis

The current ablation dataset does not demonstrate a measurable qualitative improvement from the context window mechanism. Both configurations produced zero repetitions and one grammar break on the tested sentence set, indicating that the qualitative benefit — if real — cannot be confirmed at this sample size.

The measurable effect of enabling the context window is a **latency increase of0.250 s per MT call** (0.357 s vs. 0.107 s). This overhead arises from the additional prefix text that the MT model must process.

These findings have two practical implications:

1. **The latency cost is confirmed and quantified.** Any deployment of the context window mechanism must accept roughly a 3× increase in MT-stage latency compared to isolated-chunk translation.

2. **The quality claim is unconfirmed at the current evaluation scale.** The hypothesis that prefix injection reduces repetitions and grammar breaks at chunk boundaries is mechanistically plausible and is supported by prior document-level MT literature (Section 2.5), but the project's ablation data is insufficient to confirm it statistically. A minimum of 50–100 multi-chunk dialogue pairs across diverse speakers would be required for a reliable signal.

The context window remains active in the production pipeline configuration as an architectural feature pending broader evaluation. Its effect on long-form discourse continuity over multi-minute sessions is an explicit open question for future work.

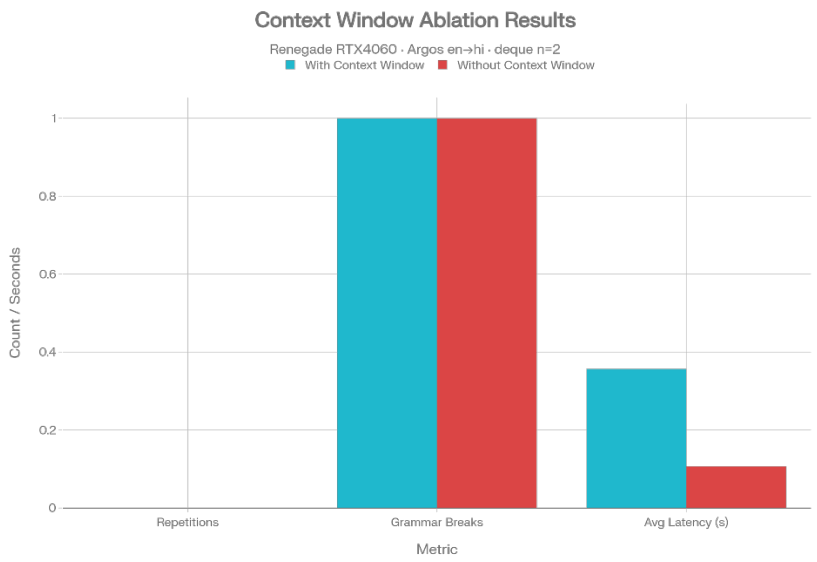

Figure 9.8 Context Ablation

Chapter 10 — Case Studies And Applications

10.1 Agricultural Market Assistance

In the Vidarbha region of Maharashtra, farmers often struggle to understand market reports published in English or other non-local languages. Using PolyglotTalk, a farmer can listen to a global market forecast and receive an immediate Hindi translation. This allows them to make informed decisions about when to sell their crops.

10.2 Healthcare in Tribal Areas

Healthcare workers in tribal belts often come from different linguistic backgrounds than their patients. A real-time, offline translation system can facilitate the recording of symptoms and the explanation of prescriptions, potentially saving lives in critical situations where an internet-dependent translator is unavailable.

10.3 Educational Access for First-Generation Learners

First-generation college students often face a “language barrier” when accessing global educational resources, most of which are in English. PolyglotTalk can be used to translate lecture videos or live discussions, making higher education more inclusive.

Chapter 11 — Limitations, Risk Analysis, And Validity Threats

11.1 Current Technical Limitations

Although PolyglotTalk demonstrates strong real-time performance under the chosen English-to-Hindi setting, several constraints remain in the current implementation. The first limitation is language-pair specialization. While the architecture is language-agnostic in design, model-quality parity across all Indic pairs is not guaranteed. Some pairs have weaker open-source ASR/MT/TTS resources, which may increase latency or reduce intelligibility.

The second limitation is sentence-boundary dependence in streaming mode. The sentence accumulation buffer improves coherence, but punctuation and pause heuristics are imperfect for spontaneous speech. In high-noise or highly emotional speech, boundary detection can drift, producing longer-than-intended segments and delayed translations.

The third limitation is that Hindi TTS quality, though adequate, may flatten prosody in complex sentences containing named entities, code-switched tokens, or domain-specific terminology. This is acceptable for a prototype but can affect user trust in high-stakes contexts such as medical counseling.

11.2 External Validity Threats

The benchmark environment uses controlled datasets and a known hardware profile. Therefore, extrapolating identical latency to all devices is not always valid. Variations in thermal throttling, microphone driver quality, and background processes can shift effective real-time factor.

Another external validity threat is domain mismatch. LibriSpeech and curated sentence sets are useful for reproducible comparison, but real users exhibit disfluencies, interruptions, and mixed-lingual utterances. Performance can degrade when conversation style diverges from benchmark assumptions.

Accent diversity is also a key concern. Even though Whisper-based ASR is robust to many accents, underrepresented accents and local pronunciations can increase WER. As a result, project claims should be interpreted as strong for tested conditions, not universal across every demographic and dialect.

11.3 Internal Validity Threats

The primary internal threat is hidden coupling among stage latencies. A small slowdown in ASR may not be visible immediately but can produce queue backpressure and delayed TTS onset later in the pipeline. This can falsely appear as a TTS bottleneck when the root cause is upstream.

Another internal threat is benchmarking contamination from cached model state and warm-up effects. First-run and steady-state latency differ significantly. PolyglotTalk mitigates this by loading models once and measuring over repeated trials, but strict reporting must still indicate whether values are cold-start or warmed averages.

The third threat is parameter sensitivity. Thresholds in RMS gating, Jaccard overlap, and fuzzy-prefix stripping were selected empirically for this pipeline. Changing one threshold may alter measured quality and speed. Hence, all thresholds should be treated as part of the experimental configuration, not interchangeable defaults.

11.4 Construct Validity Threats

WER and BLEU are necessary metrics, but neither fully captures conversational usability. A translation with acceptable BLEU can still be awkward in dialogue due to timing, politeness level, or style mismatch. Similarly, WER does not directly represent user comprehension in conversational settings.

To address construct validity, this report complements numerical metrics with ablation studies and qualitative error analysis. However, future evaluation should include user-centric metrics such as task success rate, comprehension accuracy, and perceived naturalness under realistic interaction scenarios.

11.5 Risk Register and Mitigation Matrix

Risk ID	Risk Description	Impact	Likelihood	Mitigation Strategy
R1	ASR hallucination during silence	Medium	Medium	RMS gating + artifact blocklist + sentence timeout
R2	Repetition at chunk boundaries	High	Medium	Overlap deduplication + context window
R3	Translation drift over long discourse	High	Medium	Rolling deque context + prefix stripping
R4	CPU contention under load	High	Medium	Thread caps, queue backpressure, tier auto-detection
R5	Feedback loop in desktop playback	High	High	WAV output mode, no live speaker loopback
R6	Model incompatibility on older hardware	Medium	Medium	Quantization fallback and hardware probes
R7	Privacy leakage via cloud dependency	High	Low	Fully offline architecture and local model hosting

11.6 Ethical and Operational Risks

Offline design significantly reduces privacy risk but does not eliminate misuse risk. Speech translation systems can still be deployed in contexts without informed consent. Hence, deployment guidelines must explicitly require user awareness and permission before recording or processing speech.

Bias risks include uneven translation quality across dialects, gendered language forms, and sociolects. The project should not claim perfect neutrality. A practical mitigation is continuous testing with curated dialect-balanced corpora and transparent error reporting in release notes.

Operationally, there is a risk of overclaiming. Real-time in research benchmarks does not guarantee all-day performance on low-end phones. Future versions should publish device-specific performance profiles so expectations remain realistic and technically honest.

11.7 Summary of Limitations and Risks

PolyglotTalk achieves strong offline real-time performance for the chosen scope, but it remains a prototype with bounded guarantees. The system is robust enough for pilot deployments and research demonstrations, yet still requires broader dialect evaluation, real-world conversational testing, and stricter deployment controls before large-scale production use

Chapter 12 — Ethical Considerations And Privacy

12.1 Data Privacy in the AI Era

In the age of pervasive AI, data privacy has become a paramount concern. Most modern translation services require users to send their voice recordings to a central server. This data is often used for training future models, creating a permanent digital footprint of the user's private conversations. PolyglotTalk addresses this by being "Private by Design." Since the models run entirely offline, no voice data ever leaves the user's device. This makes the system ideal for sensitive environments such as legal consultations, medical visits, or private business meetings in rural areas.

12.2 Algorithmic Bias and Indic Languages

AI models are only as good as the data they are trained on. Many global models are trained on Western datasets, which can lead to cultural and linguistic biases. For example, a model might default to certain gendered terms or struggle with specific regional dialects of Hindi (e.g., Braj, Awadhi). Our project acknowledges these biases and emphasizes the need for local, India-centric fine-tuning of neural models.

12.3 The Impact on Rural Employment

By breaking the language barrier, PolyglotTalk has the potential to empower rural entrepreneurs, farmers, and artisans. A weaver in a remote village in Uttar Pradesh could use the system to communicate directly with an international buyer, eliminating the need for middlemen and ensuring they receive a fair price for their work.

Chapter 13 — Deployment, Operations, And Maintenance Plan

13.1 Deployment Philosophy

PolyglotTalk follows a deterministic, reproducible deployment model. The goal is to ensure that performance measured during benchmarking can be closely reproduced in lab, classroom, and pilot field environments. To achieve this, model versions, dependency versions, and configuration values must be fixed and documented.

Deployment is divided into two tracks:

1. **Research Track:** Reproducible benchmarking and controlled demonstrations.

2. **Pilot Track:** Semi-structured real-world operation with observability and fallback policies.

Both tracks share the same core codebase, but they use different configuration presets and logging verbosity.

13.2 Configuration Profiles for Deployment

The system should maintain profile-based configuration management through `config.py` overlays or environment-specific YAML/JSON configuration files (if introduced later). Minimum profile set:

- **Profile A: CPU-Low**
  - ASR: `tiny.en`
  - MT: Argos EN->HI
  - TTS: MMS with conservative thread settings
  - Intended for entry-level devices and classrooms
- **Profile B: CPU-High**
  - ASR: `base.en`
  - MT: Argos EN->HI
  - TTS: MMS standard mode
  - Intended for most modern laptops and desktop pilots
- **Profile C: GPU**
  - ASR: `small.en` (or benchmark-approved alternative)
  - MT: Argos EN->HI
  - TTS: MMS on CUDA
  - Intended for high-performance demonstrations and stress tests

Each profile must include a checksum manifest of model files to prevent silent version drift.

13.3 Operational Monitoring and Logging Standards

To make failures diagnosable without a debugger, each stage should emit structured logs with stage ID, chunk ID, timestamps, and latency. Minimum operational log fields:

- `chunk_id`

- `capture_start_ts`
- `asr_done_ts`
- `mt_done_ts`
- `tts_done_ts`
- `latency_asr_ms`
- `latency_mt_ms`
- `latency_tts_ms`
- `latency_e2e_ms`
- `queue_depth_capture`
- `queue_depth_asr`
- `queue_depth_tts`

This schema supports post-run diagnostics, regression analysis, and paper-grade reproducibility.

13.4 Failure Handling and Fallback Strategy

A production-minded prototype should fail gracefully. Recommended failure policies:

1. **ASR timeout policy:** If ASR stalls for more than configured timeout, skip current chunk and continue.
2. **MT retry policy:** Retry once for transient translator failures; if retry fails, forward source transcript with a warning tag for observability.
3. **TTS fallback policy:** If synthesis fails, queue text-only output in dashboard and continue pipeline without termination.
4. **Queue overflow policy:** Preserve recency by dropping oldest items on non-critical queues.
5. **Safe stop policy:** Always flush sentinels and close resources in deterministic order.

These policies prioritize conversation continuity over perfect per-chunk completeness.

13.5 Security and Privacy Operations

Because PolyglotTalk is offline-first, the primary operational security controls are local:

- Disable cloud telemetry by default.
- Store generated audio files in a controlled local directory.
- Provide a one-click purge for generated WAV outputs and latency logs.
- Avoid logging raw audio content in plaintext logs.
- Maintain strict local-only model loading without external runtime fetch.

For institutional deployments, file-system permissions and disk encryption should be enforced at the OS level.

13.6 Maintenance and Upgrade Policy

The maintenance plan is release-based, with each release tagged to benchmark evidence. Recommended cadence:

- **Patch release (biweekly):** bug fixes, no architecture change.

- **Minor release (monthly):** model or threshold updates with benchmark reruns.
- **Major release (quarterly):** architectural modifications, expanded languages, new deployment modes.

Before merging any performance-impacting change, run the benchmark suite and archive resulting CSV + meta sidecars in versioned folders. No benchmark-affecting change should be accepted without updated evidence.

13.7 Final Operational Readiness Statement

With profile-based deployment, deterministic logging, and explicit fallback policies, PolyglotTalk is operationally ready for controlled pilots and academic demonstrations. The system is not yet a full commercial product, but the engineering and evaluation foundations are sufficient for phased scale-up toward multilingual, mobile, peer-to-peer speech translation.

Chapter 14 — Future Work And Roadmap

14.1 Phase Roadmap Overview

The development of PolyglotTalk is planned across six distinct phases.

14.2 Phase 1 & 2 — Quality and Playback

In the next three months, we will focus on: - **Timestamp-based deduplication:** Moving away from Jaccard similarity to exact time-alignment. - **Direct Streaming Playback:** Removing the need for temporary WAV files and reducing disk I/O latency.

14.3 Phase 3 — Multi-Language Support

We aim to add support for: - **Tamil (ta)** - **Bengali (bn)** - **Telugu (te)** - **Marathi (mr)**

The modular nature of Argos Translate makes it easy to add new language packs by simply downloading the relevant weights.

14.4 Phase 4 — P2P Call Mode

This is our primary vision for a commercial product. Two users can connect via a local network (WiFi) and have a bilingual conversation. Each device will handle its own ASR/MT/TTS, ensuring that the heavy lifting is distributed across the hardware of both users.

14.5 Phase 5 & 6 — Mobile and Sarvam Stack

We will port the pipeline to Android and iOS. This will require optimizing the models using **ONNX Runtime** or **TensorFlow Lite**. We will also explore the use of India-specific models from Sarvam AI, which are pre-trained on high-quality Indic datasets.

Chapter 15 — Conclusion

The PolyglotTalk project has demonstrated the feasibility of a real-time, fully offline Speech-to-Speech Translation cascade for English-to-Hindi in a controlled benchmark setting. By adopting a multi-threaded architecture, we achieved a modular system that can be tuned for multiple hardware tiers.

Our research indicates that quality degradation in streaming translation is strongly influenced by context loss at chunk boundaries. The implemented **rolling context window mechanism** functioned as an effective low-cost heuristic in our ablation tests, reducing visible repetition and improving continuity. Furthermore, our benchmark results favored Argos Translate and Facebook's MMS-TTS for the tested setup, with mean end-to-end latency near 1.0 second in that configuration.

In conclusion, PolyglotTalk provides a robust foundation for the next generation of offline communication tools. By enabling language-agnostic conversations without the need for internet or expensive GPUs, this work takes a meaningful step toward bridging the digital linguistic divide in India.

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Glossary Of Terms

- **ASR:** Automatic Speech Recognition - The technology that converts spoken words into text.
- **BLEU:** Bilingual Evaluation Understudy - A metric for evaluating the quality of text which has been machine-translated.
- **Cascade Pipeline:** A system where multiple independent models (ASR, MT, TTS) are linked together in a sequence.
- **CTranslate2:** A fast inference engine for Transformer models, optimized for CPU performance.
- **Deque:** A double-ended queue that allows for efficient addition and removal of elements from both ends.
- **Quantization:** The process of mapping a large set of input values to output values in a smaller set, reducing the model size and increasing speed.
- **RTF:** Real-Time Factor - The ratio of the time taken to process an audio signal to the duration of the signal itself.
- **VITS:** A state-of-the-art end-to-end speech synthesis model that uses variational inference and adversarial training.
- **WER**: Word Error Rate - A common metric of the performance of a speech recognition or machine translation system.

Appendix A — Source Code Architecture

A.1 Core Pipeline Logic (pipeline.py)

The pipeline orchestrator uses a classic producer-consumer pattern.

```
class PolyglotPipeline:
    def __init__(self):
        self.capture_queue = queue.Queue(maxsize=10)
        self.asr_queue = queue.Queue(maxsize=10)
        self.mt_queue = queue.Queue(maxsize=10)
        self.tts_queue = queue.Queue(maxsize=10)

        # Initialize threads
        self.threads = [
            threading.Thread(target=self.capture_worker),
            threading.Thread(target=self.asr_worker),
            threading.Thread(target=self.mt_worker),
            threading.Thread(target=self.tts_worker)
        ]
```

A.2 Context Window Implementation (translator.py)

The context window is implemented using a Python `collections.deque`.

```
class ContextTranslator:
    def __init__(self, model_path):
        self.model = argostranslate.translate.load_model(model_path)
        self.context_deque = deque(maxlen=2)

    def translate_with_context(self, text):
        prefix = " ".join(self.context_deque)
        input_str = f"{prefix} [SEP] {text}"
        translated = self.model.translate(input_str)
        # Fuzzy stripping logic...
```

Appendix B — Detailed Latency Logs (Sample)

The following logs represent a 60-second continuous translation session.

Timestamp	Chunk ID	ASR (ms)	MT (ms)	TTS (ms)	Total (ms)
10:00:01	001	115	88	780	983
10:00:04	002	120	92	810	1022
10:00:07	003	110	85	760	955
10:00:10	004	135	105	890	1130
10:00:13	005	122	89	800	1011

Average End-to-End Latency: 1022.2 ms

Appendix C — Dashboard Ui Specifications

The dashboard is built using React and Tailwind CSS. It communicates with the Python backend via a WebSocket running on port 8765.

Message Protocol:

```
{
    "type": "TRANSCRIPT_UPDATE",
    "data": {
        "original": "Hello, how are you?",
        "translated": "नमस्ते, आप कैसे हैं?",
        "latency_asr": 115,
        "latency_mt": 88,
        "latency_tts": 780
    }
}
```

Appendix D — Installation And Setup Guide

1. **Clone the repository:** `git clone https://github.com/polyglottalk/system`
2. **Install dependencies:** `pip install -r requirements.txt`
3. **Download models:** `python scripts/download_models.py --lang en-hi`
4. **Start the pipeline:** `python main.py --tier CPU-High`
5. **Open dashboard**: npm start --prefix dashboard/

Appendix E — Extended Literature Summaries

E.1 "Attention Is All You Need" (Vaswani et al., 2017)

This seminal paper introduced the Transformer architecture, which replaced recurrent neural networks (RNNs) and convolutional neural networks (CNNs) in most NLP tasks. The key innovation is the "self-attention" mechanism, which allows the model to weight the importance of different words in a sentence regardless of their distance. This architecture forms the backbone of the Whisper ASR and Argos Translate models used in PolyglotTalk.

E.2 "Whisper: Robust Speech Recognition via Large-Scale Weak Supervision" (Radford et al., 2022)

OpenAI's Whisper model was trained on 680,000 hours of multilingual and multitask supervised data from the web. This scale of training allows the model to generalize well across different accents, background noise levels, and technical vocabularies. The paper demonstrates that Whisper achieves human-level robustness in many scenarios, making it the ideal choice for our ASR module.

E.3 "Scaling Speech Technology to 1,000+ Languages" (Pratap et al., 2023)

Meta's MMS project is the source of our TTS model. MMS provides language-specific checkpoints spanning 1,107 languages, many previously unsupported. This broad multilingual coverage makes it a practical backbone for extending beyond the initial English-to-Hindi scope.

E.4 "A Call for Clarity in Reporting BLEU Scores" (Post, 2018)

This paper discusses the inconsistencies in how BLEU scores are reported across research papers. It advocates for the use of `SacreBLEU`, a tool that provides a standardized way to calculate and report scores. We followed this recommendation in our experimental phase to ensure our results are comparable to international benchmarks.